

Trend Following Strategies: A Practical Guide

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Abstract

Trend-following strategies, rooted in time-series momentum, have demonstrated enduring efficacy across diverse asset classes and market conditions. This paper provides a comprehensive exploration of the theoretical foundations, risk-return dynamics, and practical implementation of such strategies. Key findings reveal that trend following strategies exhibit convex return profiles with “crisis alpha” properties, thriving during market dislocations while offering diversification benefits. The choice of time scale for trend calculation is paramount, as strategies must align with an asset’s return characteristics. Empirical simulations and comparative analyses of trend measures underscore that methodology matters less than selecting an appropriate time horizon.

Empirical evidence from China’s futures markets (1999–2019) confirms the global applicability of trend following, with a combined multi-scale strategy achieving a 16.24% annualized return and a Sharpe ratio of 0.88, significantly outperforming buy-and-hold approaches. However, challenges persist, including periods of underperformance linked to heightened asset correlations, reduced market trends, and leverage sensitivity. Critically, the strategy’s success hinges on disciplined risk management and acceptance of short-term drawdowns. By synthesizing academic insights and practical considerations, this study reaffirms trend following as a resilient, albeit imperfect, tool for portfolio diversification and crisis hedging, urging practitioners to prioritize time-scale alignment and robust risk controls.

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1 Introduction

In quantitative investment, it is often said that the trend is your friend. Trend following is perhaps the most easily understood quantitative investment strategy. When the price of an asset rises, it is in an upward trend, and one might consider buying long; when the price of an asset falls, it is in a downward trend, and one might consider selling short. Therefore, trend following is essentially a *time-series* momentum strategy, focusing solely on the historical performance of the asset itself, without considering the performance of other assets at the current time. In contrast, another concept is *cross-sectional* momentum, with classic research from Jegadeesh and Titman (1993). At a given time t , one computes the performance of all assets over a past period, and goes long on a basket of relatively strong performing assets (winners) and goes short a basket of relatively weak performing assets (losers). Time-series momentum is an absolute concept, while cross-sectional momentum is a relative concept.

When people talk about trend following, they often refer to time-series momentum. This article will mainly focus on time-series momentum. In the following discussion, we will use the terms trend following and time-series momentum interchangeably¹. For those interested in cross-sectional momentum, please refer to the related literature regarding the momentum factor in empirical asset pricing.

1.1 Trend Following is Everywhere

As a quantitative strategy, trend following is ubiquitous. Moskowitz et al. (2012) construct a time-series momentum portfolio using 58 assets including commodity futures, bond futures, currency futures, and stock index futures. They found that within a range of 1 to 12 months, returns are persistent, meaning assets that have risen in the past are more likely to rise in the future, and assets that have fallen in the past are more likely to fall in the future. From 1985 to 2009, the time-series momentum portfolio performed excellently, with significant excess returns greater than 0.

Hurst et al. (2017) expand the research scope and time period to include 29 commodities, 11 equity indices, 15 bond indices, and 12 currency pairs, with empirical data starting from 1880². When constructing the trend following portfolio, they considered three different frequencies: 1 month, 3 months, and 12 months. From 1880 to 2013, the trend following portfolio achieved an annualized return of 14.9%, and even after considering a 2% management fee and 20% performance fee, it still achieved an annualized return of 11.2%, with an annualized volatility of only 9.7%. Babu et al. (2020) add 82 alternative assets and 16 factor return series to the 58 traditional assets

¹Generally, strategies that utilize price trends for trading are referred to as trend following in industry, while the model proposed by Moskowitz et al. (2012) in academia is termed time-series momentum. These two concepts are essentially the same and often lack strict boundaries. Therefore, this article does not distinguish between them.

²They used spot prices if futures were not available.

covered by Moskowitz et al. (2012), demonstrating that trend following is widespread among these assets and remains effective in the long term. By expanding into new asset types, the Sharpe ratio of a trend following strategy can be largely improved.

These empirical findings all indicate that trend following strategies are present across time dimensions, asset categories, and investment cycles, with impressive performance.

1.2 Underlying Logic

The long-term vitality of trend following must be supported by solid logical foundations. As all history is in-sample, therefore when evaluating whether an investment strategy holds water, if one focuses too much on risk-return metrics like returns or Sharpe ratios without paying attention to why it works, even the most impressive data can be fragile.

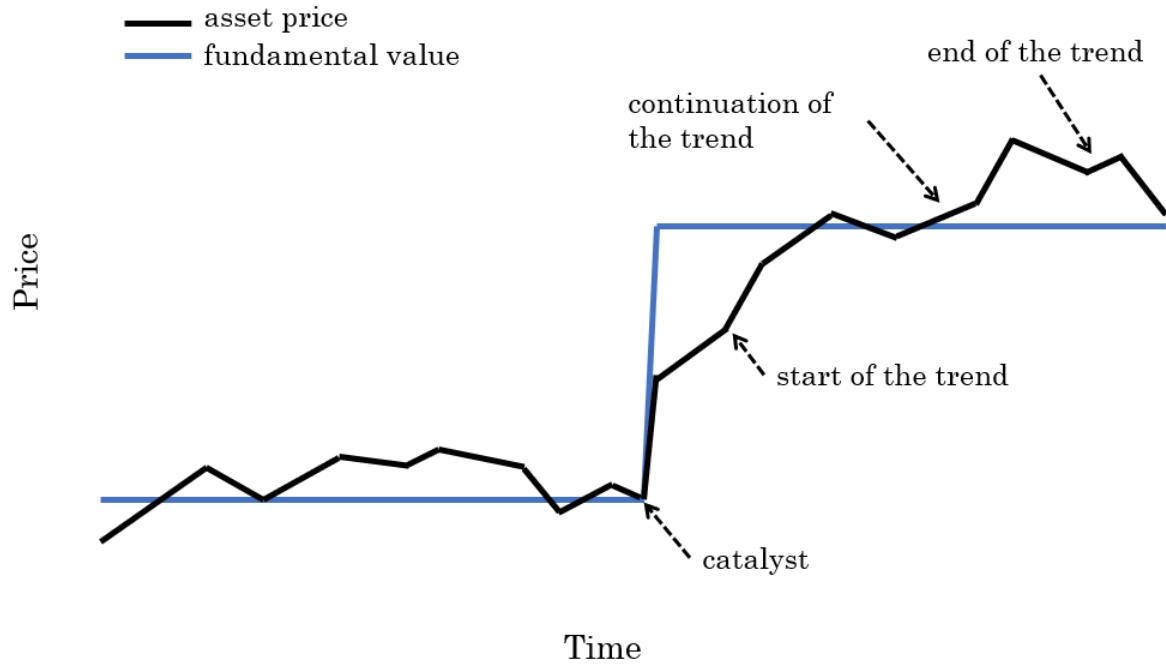
Academically, when explaining the source of factor returns, two perspectives are often considered: risk compensation and mispricing. The former is from the perspective of classical finance, suggesting that an anomaly can achieve excess returns because it bears some systematic risk, and the excess returns are compensation for that risk; the latter is from the perspective of behavioral finance, suggesting that either some constraint prevents the anomaly from disappearing, or investor cognitive biases lead to irrational behavior.

Moskowitz et al. (2012) study the source of trend following returns from a risk perspective. Time-series regression results showed that trend following cannot be explained by existing risk factors (such as HML, SMB, etc.); not only is it difficult to identify the source of risk, but the authors also find that trend following performs well when the market performs poorly, such as during the 2008 financial crisis when (Commodity Trading Advisor) CTA funds stood out. Therefore, it is difficult to understand trend following from a risk compensation perspective. It is much easier to discuss from a behavioral finance perspective, as Clare et al. (2016) and Hurst et al. (2017) have done in detail. The efficient market hypothesis suggests that when important new information appears, prices will immediately react to match the latest intrinsic value. However, due to investor cognitive biases, the market is not completely efficient but slowly absorbs the impact of new information.

Generally, the formation of a trend can be divided into three stages, as shown in Figure 1³. In the initial stage, due to anchoring effects and disposition effects, prices underreact to information.

- **Anchoring Effect:** Investors tend to anchor their views on recent historical data, unwilling to quickly change their minds, leading to slow reactions to new information and thus slow price reactions.
- **Disposition Effect:** Investors tend to sell profitable stocks too early to realize gains; conversely, they are reluctant to cut losses on losing stocks. The consequence is that prices do

³This figure is redrawn based on Hurst et al. (2017).



Source: Redrawn based on Hurst, Ooi and Pedersen (2017)

Figure 1: Trend Formation Process

not rise or fall abruptly but move slowly.

- **Non-Profit Operations:** Some non-profit operations also slow down price reactions, such as central banks intervening in the foreign exchange and fixed income markets to reduce exchange rate and interest rate volatility.

Once a trend starts, it enters the second stage: trend continuation or even overreaction, often due to herding effects, confirmation bias, and representativeness bias.

- **Herding Effect:** Also known as the bandwagon effect, when investors' views differ from the majority, they tend to doubt and change their views to align with the group. After a price rise or fall begins, investors will join the trend like a herd.
- **Confirmation and Representativeness Bias:** After a trend starts, investors use recent price movements to infer the future, selectively focusing on the direction of recent profits; moreover, once they confirm their views, they will seek various data and information to support their views, allowing the price trend to continue.

Finally, trends do not last forever, and prices do not move in one direction indefinitely. In the end, prices may have overreacted, deviating significantly from fundamentals, leading to a reversal and the end of the trend.

1.3 Outline

The rest of this paper is organized as follows. Chapter 2 discusses the risk-return characteristics of trend following strategies, introducing the concepts of divergent risk-taking and the convex nature of the return profile. Chapter 3 covers the importance of time scales for calculating trends and presents empirical simulations highlighting the effectiveness of different time scales for trend following strategies. In Chapter 4, we compare various trend calculation methods, including time-series momentum, moving average crossovers, and Kalman filtering, and explore the relative effectiveness of each. Chapter 5 examines position sizing and leverage management strategies, emphasizing target volatility approaches and their impact on portfolio risk. Chapter 6 addresses the essence of trend signals, discussing why trend following is inherently a right-side trading strategy and the causes of the signal lag. Chapter 7 explores the properties of trend following strategies, including crisis alpha, long-short asymmetry, and the win rate/profit-loss ratio. Chapter 8 presents extensive empirical evidence from China's futures markets, analyzing the performance of trend following strategies across various sectors. Chapter 9 investigates whether trend following strategies have failed and the challenges faced by practitioners. Chapter 10 discusses the critiques and controversies surrounding trend following strategies. Finally, Chapter 11 concludes.

2 Risk-Return Characteristics

2.1 Divergent Risk Taking

Jusselin et al. (2017) point out that the return characteristics of time-series momentum strategies is convex. When the trend is small, the strategy yields negative returns; only when the trend is significant (either upward or downward) does the strategy generate positive returns.

According to this analysis, the losses of time-series momentum strategies are limited, while the potential gains are unbounded. It is worth noting that the limited losses are conditional, requiring that the strategy does not employ excessive leverage (as high leverage non-linearly amplifies both gains and losses) and that asset prices are continuous. Examining the PnL curve of trend following strategies, it is often observed that such a strategy often experiences a series of small drawdowns, followed by a significant upward movement (a large trend), which offsets the previous drawdowns, resulting in positive long-term returns. This characteristic of limited losses and unlimited gains leads to the positive skewness of the strategy's return distribution. This distribution is also known as the "Holy Grail" distribution.

It is precisely this unique characteristic that allows trend following strategies to stand out when other assets experience tail risks, which is why they are renowned as crisis alpha⁴. Why do different strategies exhibit different distribution characteristics in their returns? This can be

⁴We will talk more about this in Section 7.

explained by two types of risks. Any investment strategy that aims to generate profits must bear certain risks. The type of risk a strategy assumes determines the characteristics of its return distribution. Risks can be broadly categorized into convergent risks and divergent risks, and thus strategies can also be classified into convergent risk-taking and divergent risk-taking.

- Convergent risks are those that are understood and can be measured through modeling;
- Divergent risks are unknown, unpredictable, and cannot be well-measured risks.

When constructing a convergent risk-taking strategy, people make investment decisions based on prior beliefs about the asset's price movement derived from financial or economic theory. Conversely, when constructing a divergent risk-taking strategy, people assume that the future movement of the asset is unknown and make decisions based on the asset's past price movements without using any a prior beliefs about whether the asset will rise or fall.

For example, value investing in the stock market is a representative of convergent risk-taking. This type of strategy involves in-depth research to uncover the intrinsic value of companies, buying when the price is below the intrinsic value, and patiently waiting for the price to revert to its intrinsic value, disregarding short-term fluctuations or even sustained declines. These are the risks that this type of strategy is willing and actively chooses to bear. On the other hand, trend following strategies are representatives of divergent risk-taking. These strategies do not make any judgments about the intrinsic value of the asset but strictly trade based on price movements: buying when the price rises and selling when the price falls. They gradually increase positions when profitable and quickly close positions when incurring losses. Because of this, trend following strategies belong to divergent risk-taking strategies, and their returns exhibit the characteristics of the holy grail distribution, which is why trend following strategies remain enduringly popular.

It is necessary to point out that while the return distribution of a trend following strategy is appealing, achieving such an outcome is extremely challenging. The most difficult aspect of trend following strategies is holding onto positions during trends to let profits run (as only in this way can the right-tail of the return distribution be achieved). However, due to cognitive biases, people tend to have an excessive preference for low returns with high certainty, thus often closing profitable positions prematurely. Only by overcoming these human errors can one truly enjoy the benefits of the holy grail distribution provided by trend following strategies.

2.2 Trend vs. Volatility

Assuming asset prices follow a geometric Brownian motion:

$$\frac{dP}{P} = \mu dt + \sigma dB, \quad (1)$$

where P is the asset price, μ is the drift rate, σ is the volatility, and B is a standard Brownian motion. From this model, it is easy to see that (long) trend following strategies favor positive drift rates in returns (i.e., the larger μ , the better) and dislike volatility (i.e., the smaller σ , the better). A larger μ and/or a smaller σ indicate a more significant trend⁵. Therefore, when evaluating whether an asset is suitable for trend following strategies, both μ and σ must be considered. Hence, the industry considers the Sharpe ratio an excellent indicator of the strength of an asset's trend. Similarly, for strategies that can short, we pay more attention to the absolute value of the Sharpe ratio.

Using options pricing terminology, let S represent the net value of the trend following strategy, and P and σ represent the price and volatility of the underlying asset, respectively. The Delta and Vega of the strategy are:

$$\text{Delta} = \frac{\partial S}{\partial P}, \quad \text{Vega} = \frac{\partial S}{\partial \sigma}. \quad (2)$$

Delta can be understood as the impact of changes in asset prices on the net value of the trend following strategy. Taking a long strategy that chases upward trends as an example, it should have a positive Delta, meaning that as prices rise, the strategy's net value increases. On the other hand, Vega represents the impact of return volatility on the net value of the trend following strategy.

Roncalli (2017) points out that trend following strategies have negative Vega, meaning that short-term volatility in returns causes a decline in the strategy's net value. Since "volatility is common, while trends are not," repeated volatility in assets dominates the performance of trend following strategies (slow declines, losses) until a significant trend emerges and is captured by the strategy. Taking the U.S. stock market as an example, as it reflects economic growth and the profit growth of public companies, the market exhibits a slow bull structure. In such a structure, trend following strategies that use long-term trend signals perform better than those that use short-term trend signals.

3 Time Scale for Calculating the Trend

Often, one may ask what the most important aspect is to consider for making a profit with a trend following strategy. Literature tell us that the key elements in trend following include the trend measure (i.e., selecting the best indicator to measure the strength of a trend) and position sizing (position management). Undoubtedly, these two points are crucial for trend strategies. However, in our view, the most important aspect is choosing the appropriate time scale for calculating the trend.

For example, Winton Capital conducted a study on the time scales of trend strategies for 20 assets from four major categories (Winton 2013). In their empirical study, they considered a period of 30 years since 1984, and defined trend following strategies of three different time scales

⁵Of course, for strategies that can short, only the absolute value of μ needs to be large.

— fast (short time scale), medium, and slow (long time scale) — based on the turnover rates of the strategies (higher turnover implies a shorter time scale) and trades the aforementioned 20 assets. After adjusting for risk, the performance of these three strategies can be summarized as follows. The high turnover strategy basically became ineffective starting from 2004, while the medium and slow trend following strategies remained effective after 2004. Since 2012, the performance of the medium strategy has also weakened compared to its previous performance, whereas the slow strategy has remained relatively stable, although its absolute return is not as high as that of the fast and medium turnover strategies.

Let's look at two other examples. Figure 2 shows long-only trend following strategies using the same parameters for the Shanghai Stock Exchange (SSE) Composite Index and the S&P 500 Index as assets. These parameters were tuned for the Chinese stock market via backtesting. The results show that the trend parameters suitable for the Chinese stock market are completely ineffective for the US stock market⁶.

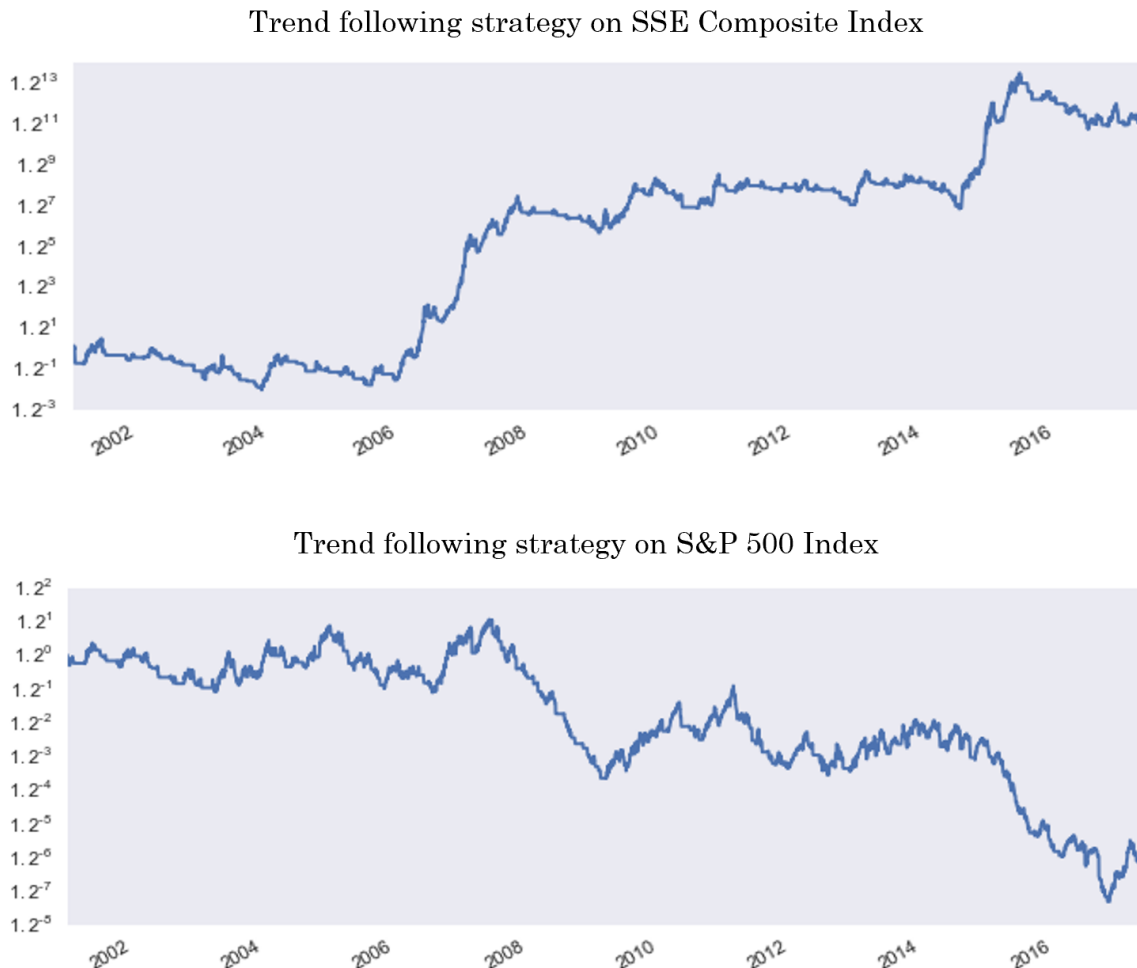


Figure 2: Examples of Trend Following Strategies

The sharp contrast illustrates that the effectiveness of a trend following strategy depends on

⁶This is not uncommon in practice.

matching the time scale for calculating the trend with the return characteristics of the asset. Next, we explore the return characteristics of assets and how that impact the performance of trend following strategies via intensive simulation studies.

3.1 Model of Log Return

The above examples show that any trend following strategy is only applicable to investment assets whose returns conform to its assumptions. So, what are the general characteristics of returns for investment assets? How can we use these characteristics to select a suitable trend following strategy? To explore this, this section decomposes returns into three parts: (long-term) mean, autocorrelation, and white noise, and studies them through simulations. Reviewing what we learned in time series analysis lectures, here are the definitions:

- **Long-term Mean:** It represents the mean of the returns per unit time over the long term.
- **Autocorrelation:** To simplify the discussion, we only consider the autocorrelation at lag 1 here; it represents the correlation of returns between two adjacent units of time.
- **White Noise:** The random fluctuation of returns within each unit of time; the random fluctuations at different times are independent of each other.

Among these three parts, white noise is the random disturbance; for the other two parts, the higher the long-term mean and the stronger the autocorrelation, the more effective the trend following strategy will be. So, which of these two factors is more important? How do they respectively affect the performance of a trend following strategy? To facilitate analysis, we use a mathematical model to artificially synthesize log returns. These log returns satisfy a predefined long-term mean μ , autocorrelation ρ , and the volatility of white noise σ . The procedure to generate the return time series is summarized below:

1. **Initialization:** $b_0 \sim \mathcal{N}(0, \sigma)$,
2. **White Noise Sequence:** $e_t \sim \mathcal{N}(0, \sigma)$ for $t = 1, 2, \dots$,
3. **Auxiliary Series Generation:** $b_t = \rho \cdot b_{t-1} + e_t$ for $t = 1, 2, \dots$,
4. **Return Series Construction:** $r_t = b_t - \mu_b + \mu$ for $t = 0, 1, 2, \dots$, where μ_b is the sample mean of b_t .

In the procedure above, e_t is a white noise series that follows a normal distribution with a mean of 0 and a standard deviation of σ . r_t is the simulated log return series. To ensure that it satisfies the given parameters μ and ρ , an auxiliary series b_t is used in the construction process. First, the series $\{b_t\}$ is iteratively constructed to satisfy the autocorrelation ρ . Since constructing

$\{b_t\}$ may introduce an unexpected mean μ_b , the process removes μ_b and adds back the given mean μ when constructing r_t . Thus, the final generated log return series $\{r_t\}$ satisfies both parameters.

Using this mathematical model, with μ , ρ , and σ as inputs, we can generate a log return series. To ensure this study is relevant to investment practice, the values of these three parameters need to be close to reality. Additionally, since the focus of the study is on the impact of μ and ρ on trend following strategies, the value of σ is fixed.

Assuming we are synthesizing a daily return series of the stock market, a reasonable value for σ is 0.01. When determining the values of μ and ρ , consider the following information as reference:

- SSE Composite Index from May 26, 2005, to May 25, 2017 has a mean of 0.037% and an autocorrelation of 0.023.
- SSE Composite Index during the bull market of 2015 (February 9, 2015, to June 12, 2015) has a mean of 0.6% and an autocorrelation of 0.09.
- SSE Composite Index after the stock crash (March 1, 2016, to May 25, 2017) has a mean of 0.049% and an autocorrelation of 0.048.
- S&P 500 Index after the global financial crisis (January 1, 2010, to January 1, 2017) has a mean of 0.04% and an autocorrelation of -0.049 .

As seen, apart from the bull market period in the Chinese stock market⁷, the mean log daily returns are around 0.04%. In terms of autocorrelation, except for the negative value of the S&P 500⁸, the autocorrelation of the Chinese stock market in different periods ranges from 0.02 to 0.1. Based on the above data, we choose the following values μ and ρ :

- μ : 0.01%, 0.05%, 0.1%.
- ρ : 0.01, 0.1, 0.2.

Combining these two parameters in pairs results in nine different combinations of μ and ρ , allowing us to construct log return series of any length using the aforementioned mathematical model. In the simulation, each series contains 2500 points⁹. Additionally, since we are generating log returns, the cumulative sum of this series represents the corresponding log price curve. In the experiment, we assume the initial price of the investment asset is 100. Figure 3 shows a set of sample price curves generated under these nine combinations of μ and ρ .

From these simulated price curves, we can observe the following:

⁷Note that the mean return during this period was an order of magnitude higher than usual, which will be discussed later.

⁸Although it is negative, it is very close to 0.

⁹Assuming 250 trading days per year, this length corresponds to ten years.

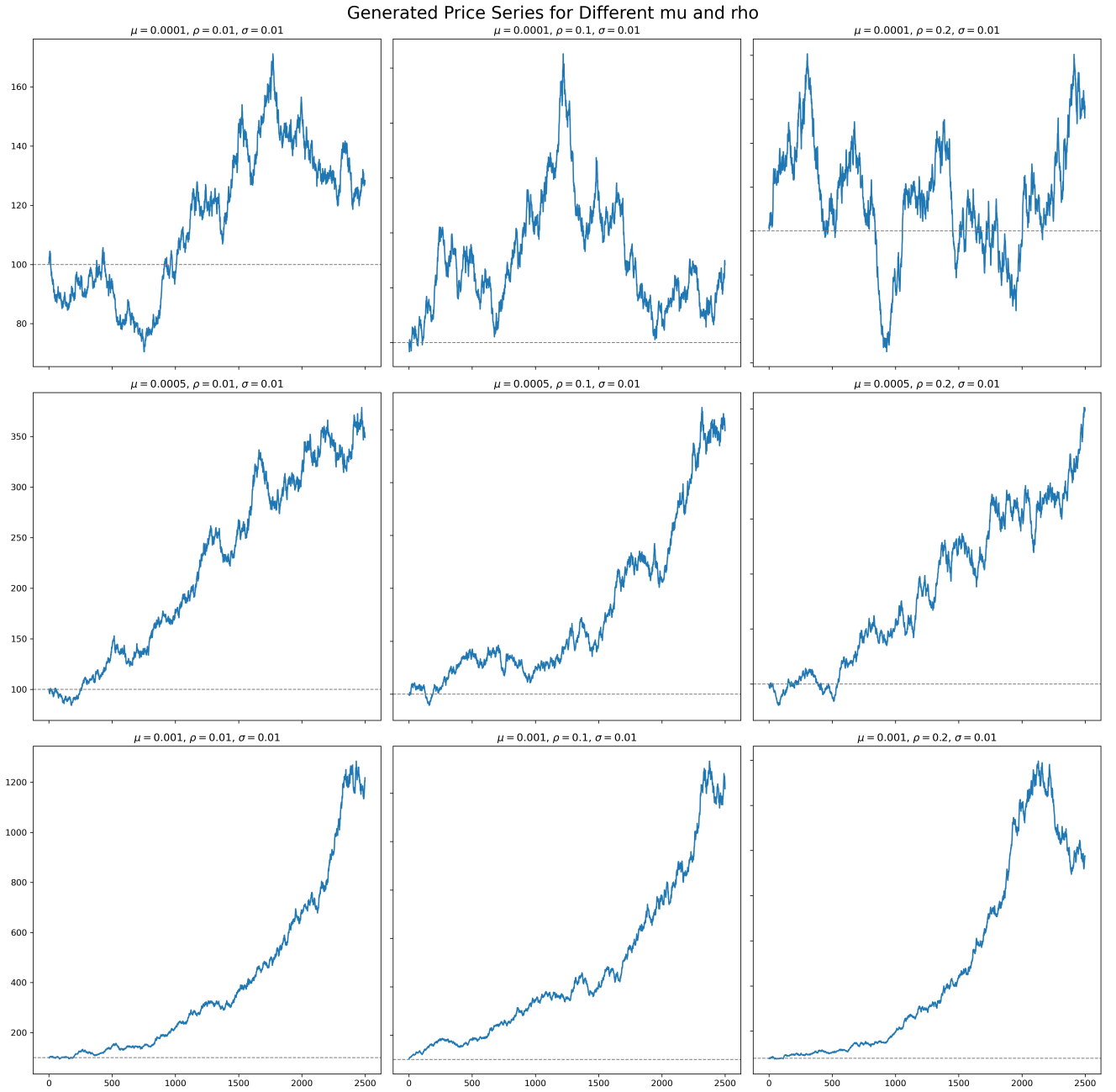


Figure 3: Examples of Generated Prices

- When μ is much smaller relative to the volatility σ (about two orders of magnitude smaller), the local trend of the price curve is dominated by autocorrelation ρ (and random noise). Since μ is too small relative to the random noise σ , no significant upward trend can be observed on any local scale.
- When μ is between one and two orders of magnitude smaller than σ (which is the typical market condition), the importance of μ to the price trend gradually increases. At this point, small autocorrelation ρ has a minimal impact on the price; when the autocorrelation ρ is large enough, it may cause local phases of rising and falling intervals.

- When μ becomes large enough to be only one order of magnitude smaller than the volatility σ or even larger (which is the case in a bull market), the influence of the mean on the price makes the impact of the autocorrelation ρ on the price negligible. The price sequence shows a continuous upward trend.

How do these different parameters affect the trend following strategy? We explore this question next.

3.2 Trend Following Performance

For each combination of μ and ρ , we generated 5000 simulations, each with 2500 data points. For each simulation, we implemented a dual moving average strategy with:

- Slow moving average: 50-day simple moving average.
- Fast moving average: 13-day simple moving average.

The trading rules are:

- When not holding any position, buy when the fast moving average crosses above the slow moving average.
- When holding a position, sell when the fast moving average crosses below the slow moving average.

Performance metrics are:

- Annualized return.
- Annualized Sharpe ratio.
- Maximum drawdown.

For each combination of μ and ρ , we plotted histograms for the distributions of these performance metrics in Figures 4 to 12.

Case 1: $\mu = 0.01\%$ and $\rho = 0.01$

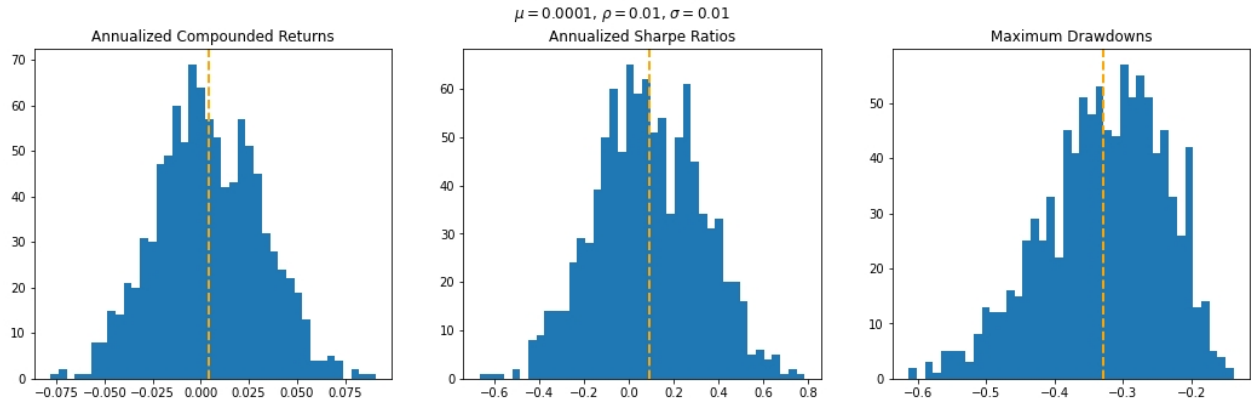


Figure 4: Simulation Performance (Case 1)

Case 2: $\mu = 0.01\%$ and $\rho = 0.1$

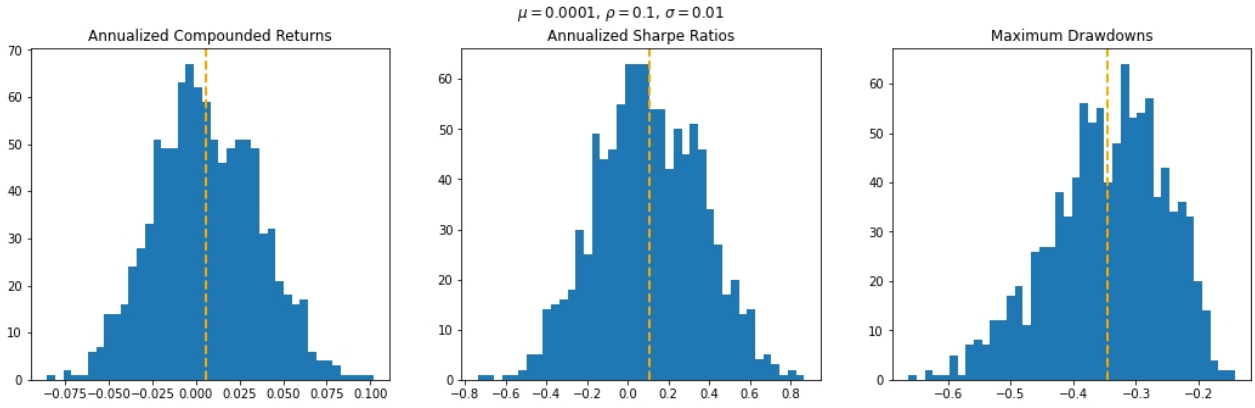


Figure 5: Simulation Performance (Case 2)

Case 3: $\mu = 0.01\%$ and $\rho = 0.2$

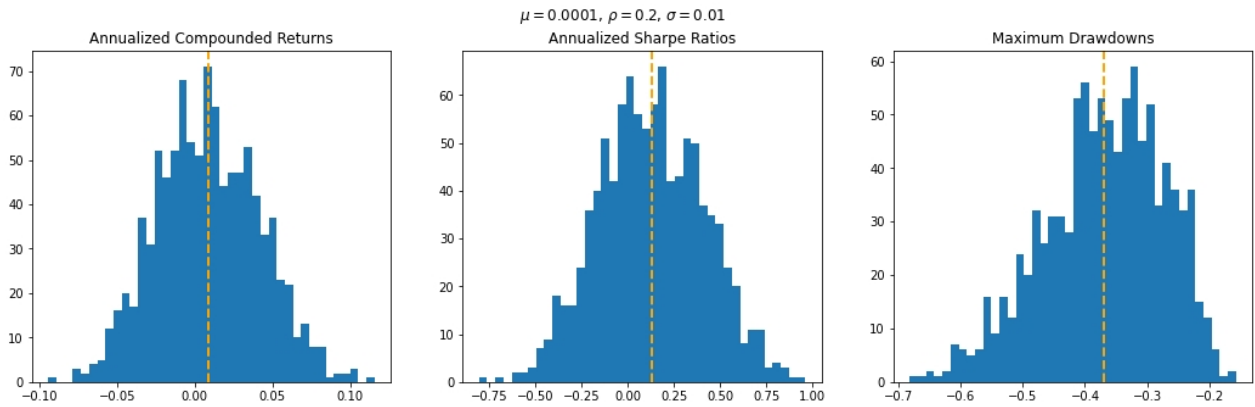


Figure 6: Simulation Performance (Case 3)

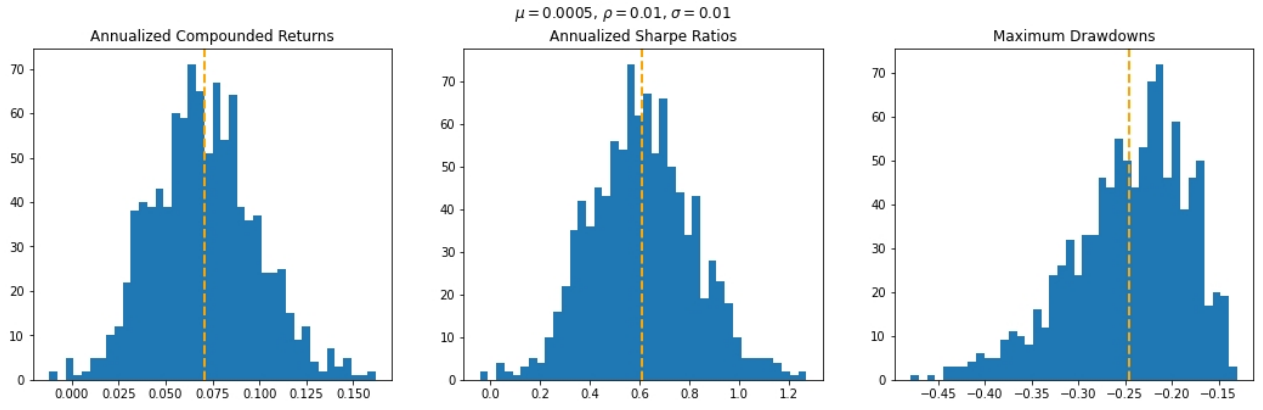
Case 4: $\mu = 0.05\%$ and $\rho = 0.01$ 

Figure 7: Simulation Performance (Case 4)

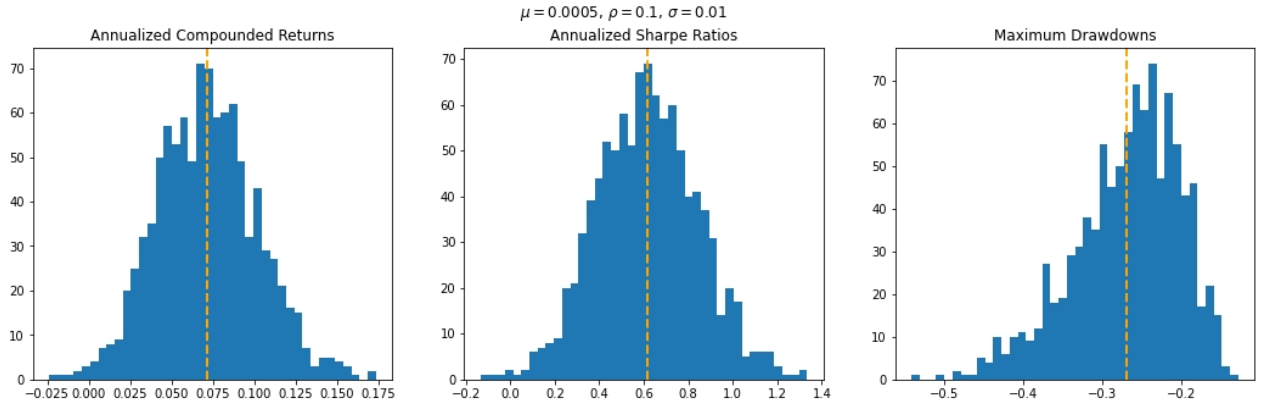
Case 5: $\mu = 0.05\%$ and $\rho = 0.1$ 

Figure 8: Simulation Performance (Case 5)

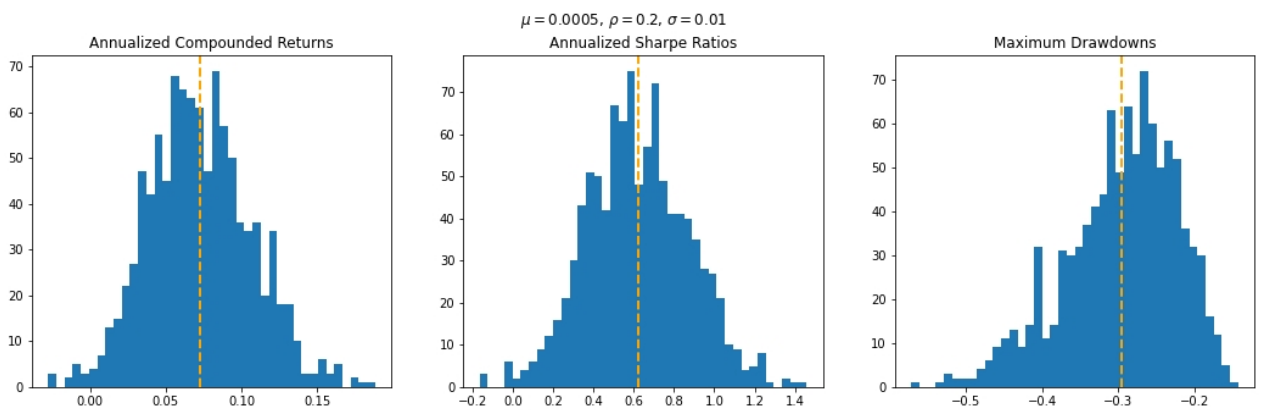
Case 6: $\mu = 0.05\%$ and $\rho = 0.2$ 

Figure 9: Simulation Performance (Case 6)

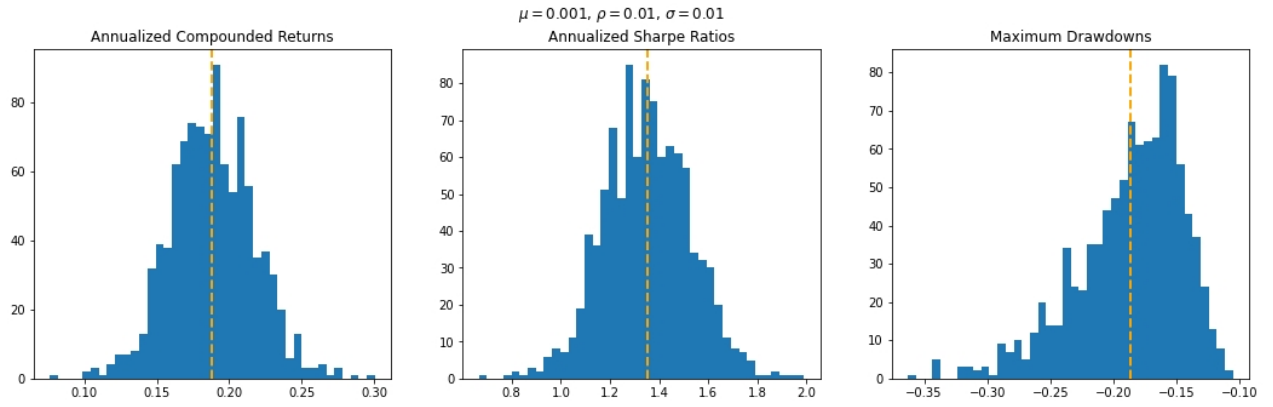
Case 7: $\mu = 0.1\%$ and $\rho = 0.01$ 

Figure 10: Simulation Performance (Case 7)

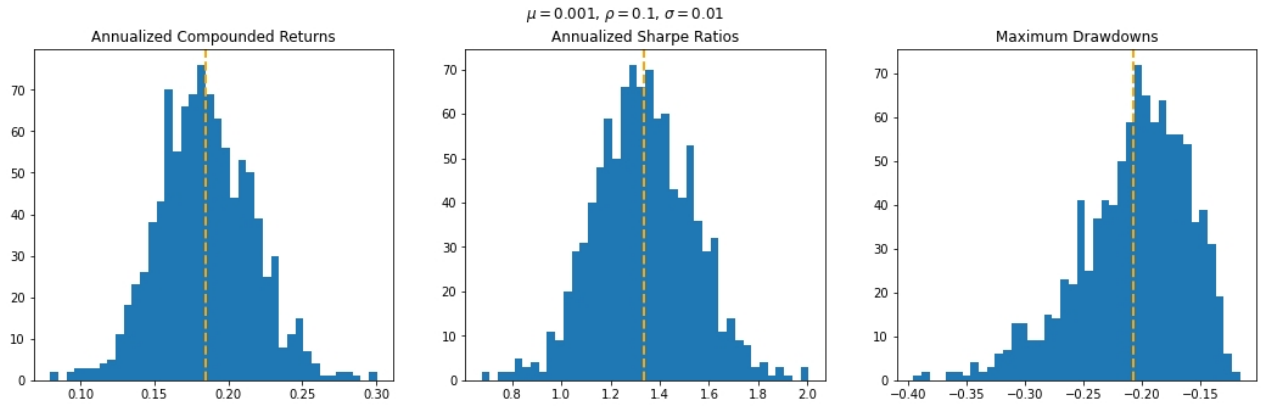
Case 8: $\mu = 0.1\%$ and $\rho = 0.1$ 

Figure 11: Simulation Performance (Case 8)

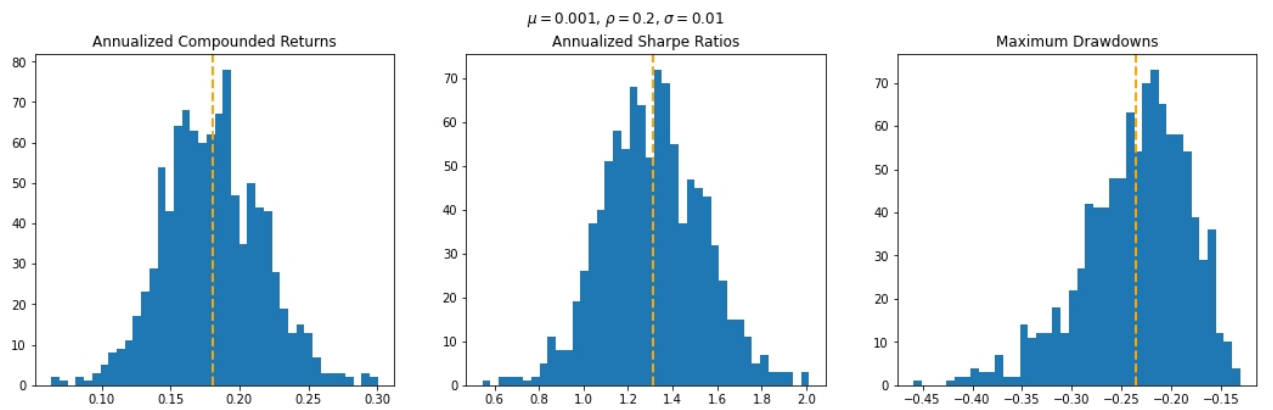
Case 9: $\mu = 0.1\%$ and $\rho = 0.2$ 

Figure 12: Simulation Performance (Case 9)

We observe the following from the experimental results for these nine sets of parameters:

1. In the first to third sets of parameters, since the long-term mean is two orders of magnitude smaller than the random noise, the price curves show frequent fluctuations across different time scales in the long term. As a result, the strategy frequently generates false signals. The average performance of the strategy under these three sets of parameters fails to achieve positive returns.
2. In the fourth to sixth sets of parameters, as the long-term mean increases, the combined effect of the mean and autocorrelation allows the price curves to overcome random noise and exhibit upward trends across different time scales. When the scale of the upward trend matches the parameter settings of the dual moving average strategy, the strategy achieves positive returns.
3. In the seventh to ninth sets of parameters, due to the further increase in the long-term mean, the effectiveness of the trend strategy improves. The expected performance (i.e., expected Sharpe ratio and expected return) of the trend following strategy is far from zero.

From these results, we can draw the following conclusions:

1. μ dominates over autocorrelation ρ — compared to autocorrelation, the low-frequency component μ represents the real trend that can be captured by a trend following strategy.
2. The difference in the orders of magnitude between μ and σ affects the performance of the strategy, which implies that choosing an appropriate time scale for calculating the trend is crucial.

4 Which Trend Measure?

On the technical side, there are many ways to measure trends: time series momentum based on past cumulative returns, moving average crossovers or channel breakouts from technical analysis, various state space models like Kalman filter, linear regression methods using prices, and even frequency domain analysis (such as Fourier transform, wavelet analysis, etc.).

When the backtest results of a particular method are not promising, people's first reaction is always to look for another (and usually more complex) method to calculate the trend. However, at their core, all the above methods are based on the information contained in the time series of prices or returns. Is there really a method or indicator that significantly outperforms others? This is the question that concerns this section.

4.1 Trend Signature Plots

To answer this question, Levine and Pedersen (2016) introduce the concept of trend signature plots. The purpose is to plot different trend calculation methods as functions of the weights of prices or returns over time steps. This allows us to compare different trend measures. Next, we explain trend signature plots using time series momentum and moving average crossovers as examples.

To facilitate discussion and mathematical derivation, we consider log prices (denoted by p_t for time t) — because the difference of log prices is the log return (i.e., $r_t = p_t - p_{t-1}$), which allows for a straightforward comparison between prices and returns when evaluating different trend calculation methods.

Time series momentum is generally calculated by taking a weighted average of returns over a past period (commonly using equal weights), resulting in a trend indicator. Assuming the current time is t , the time series momentum (TSMOM) indicator at time t is:

$$\text{TSMOM}_t = \sum_{s=1}^{\infty} c_s (p_{t-s+1} - p_{t-s}). \quad (3)$$

For generality, the above expression is written as an infinite series. In practical applications, for a given calculation window N , the weights c_s for all $s > N$ are set to zero.

Now, let's look at the moving average crossover. It uses two moving averages — a fast and a slow one — to form the trend indicator. Each moving average is a weighted average of the price series over a given window (with different weights resulting in simple moving averages, exponential moving averages, etc.). The trend signal is the difference between the fast and slow moving averages. The mathematical expression for the moving average crossover is:

$$\begin{aligned} \text{MA}_t^{\text{fast}} &= \sum_{s=1}^{\infty} w_s^{\text{fast}} p_{t-s+1}, \\ \text{MA}_t^{\text{slow}} &= \sum_{s=1}^{\infty} w_s^{\text{slow}} p_{t-s+1}, \\ \text{MACROSS}_t &= \text{MA}_t^{\text{fast}} - \text{MA}_t^{\text{slow}} \\ &= \sum_{s=1}^{\infty} (w_s^{\text{fast}} - w_s^{\text{slow}}) p_{t-s+1}. \end{aligned} \quad (4)$$

Comparing TSMOM and MACROSS methods for calculating trends, they may appear very different. However, through simple derivation, we can see that MACROSS can be easily expressed as a combination of returns weighted by a specific set of weights. Specifically, starting from the

definition of TSMOM, we can transform it into the following form:

$$\begin{aligned}
 \text{TSMOM}_t &= \sum_{s=1}^{\infty} c_s (p_{t-s+1} - p_{t-s}) \\
 &= c_1 (p_t - p_{t-1}) + c_2 (p_{t-1} - p_{t-2}) + \cdots \\
 &= c_1 p_t + (c_2 - c_1) p_{t-1} + (c_3 - c_2) p_{t-2} + \cdots .
 \end{aligned} \tag{5}$$

Comparing the above expression with the definition of MACROSS, we can see the relationship between the weights c_s and the MACROSS weights w_s^{fast} and w_s^{slow} :

$$\begin{aligned}
 c_1 &= w_1^{\text{fast}} - w_1^{\text{slow}} \\
 c_s &= c_{s-1} + w_s^{\text{fast}} - w_s^{\text{slow}} \\
 &= \sum_{j=1}^s (w_j^{\text{fast}} - w_j^{\text{slow}}), \quad s \geq 2.
 \end{aligned} \tag{6}$$

This result implies that if we weight the returns by c_s , the resulting trend signal is equivalent to the moving average crossover. Using this equivalence, different trend calculation methods can be plotted as the weights of returns over time, resulting in a trend signature plot. It is important to note that the sum of the return weight series obtained by converting the price weights does not equal 1. To compare the trend signature plots of different methods, the return weights need to be normalized.

The above derivation for moving average crossovers can also be generalized to more common cases. Suppose a trend calculation method is a weighted average of the price series: $\sum w_i p_i$, then the corresponding return weight coefficient is:

$$c_s = A \sum_{j=1}^s w_j, \quad s = 1, 2, \dots, \tag{7}$$

where A is the normalization constant for $\{c_s\}$. Figure 13 shows the trend signature plots for three different methods (with the horizontal axis starting from $s = 0$ representing the most recent moment and increasing to represent larger time lags):

1. TSMOM signal with a window of 260 trading days and equal weighting.
2. Moving average crossover signal calculated with simple moving averages over windows of 20 and 260 days.
3. Moving average crossover signal calculated with exponential moving averages with centers of mass of 32 and 128.

From the figure, it is evident that despite the apparent differences between time series momentum and moving average crossovers, their underlying trend signatures are quite similar; the

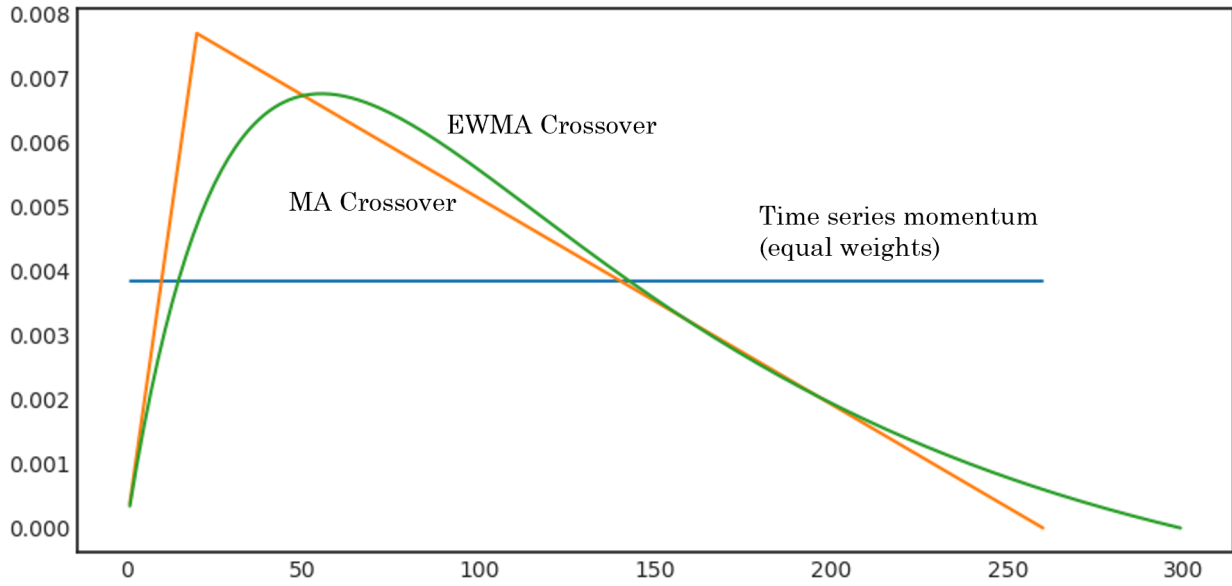


Figure 13: Trend Signature Plot 1

similarity is even higher between simple moving averages and exponential moving averages, which are both forms of moving average crossovers. This indicates that it is unrealistic to expect exponential moving averages to largely outperform simple moving averages.

4.2 Other Measures

This section considers two other common methods to measure trend: Kalman filter and linear regression.

Kalman filter (Kalman 1960) is an effective method for estimating the hidden state from a noisy dynamic linear system. For a price series, the simplest model is the random walk plus local trend, where the value of the trend is unobservable and needs to be estimated using Kalman filter. Harvey (1984) points out that when using Kalman filter to analyze the local trend model, the optimal estimate of the local trend is the exponential moving average of the return series.

Figure 14 compares the trend signature plots of three different trend calculation methods:

1. TSMOM signal with a window of 260 trading days and equal weighting;
2. Moving average crossover signal calculated with exponential moving averages with centers of mass of 32 and 128;
3. Kalman filter trend signal represented by exponential smoothing with a center of mass of 96.

Compared to the other two methods, Kalman filter tends to assign higher weights to recent returns. Other common linear filters, such as the Hodrick and Prescott (1997) filter, can yield similar results, which will not be elaborated here.

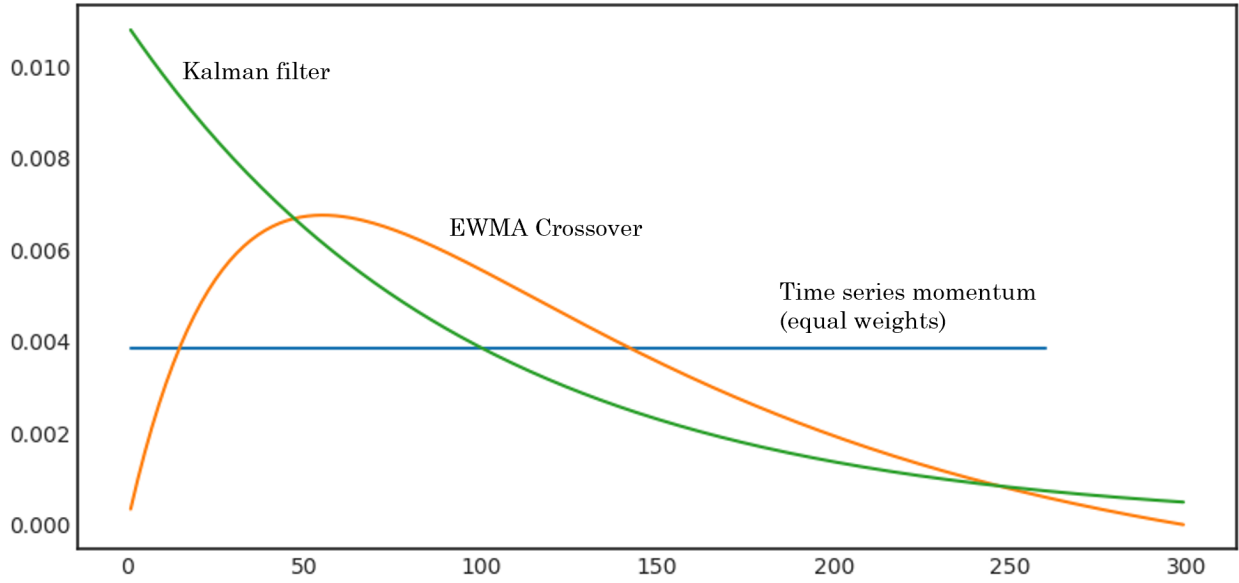


Figure 14: Trend Signature Plot 2

Now, let's look at estimating a linear regression model using OLS. In particular, we regression the log price p_s to the time index s , i.e., $p_s = \alpha + \beta s + e$, and use the sign of the regression coefficient β as an indicator of an upward or downward trend. Specifically, assume the window for regression is N and therefore the sample points are $(1, p_{t-N+1})$, $(2, p_{t-N+2})$, $\dots$, $(N-1, p_{t-1})$, (N, p_t) . The OLS estimator of β is:

$$\hat{\beta} = \frac{\sum_{s=1}^N (p_{t-s+1} - \bar{p}) \left(\frac{N+1}{2} - s \right)}{\sum_{m=1}^N \left(\frac{N+1}{2} - m \right)^2}, \quad (8)$$

where $\bar{p}$ is the average price over N periods. Additionally, through simple algebra, we can find:

$$\sum_{s=1}^N \bar{p} \left(\frac{N+1}{2} - s \right) = \bar{p} \left[N \left(\frac{N+1}{2} \right) - \sum_{s=1}^N s \right] = \bar{p} \left[N \left(\frac{N+1}{2} \right) - N \left(\frac{N+1}{2} \right) \right] = 0. \quad (9)$$

Thus, the expression of $\hat{\beta}$ can be further simplified to:

$$\hat{\beta} = \sum_{s=1}^N \left[\frac{\left(\frac{N+1}{2} - s \right)}{\sum_{m=1}^N \left(\frac{N+1}{2} - m \right)^2} \right] p_{t-s+1}. \quad (10)$$

This implies that $\hat{\beta}$ is a weighted average of the prices. Therefore, we can use the method mentioned in the previous section to convert it into weighted average of returns and obtain the trend signature plot. Figure 15 compares the trend signature plots of three different trend calculation methods, showing that the trend signals calculated by OLS are essentially no different from TSMOM and MACROSS:

1. TSMOM signal with a window of 260 trading days and equal weighting;
2. Moving average crossover signal calculated with simple moving averages over windows of 20 and 260 days;
3. Trend signal obtained by linear regression.

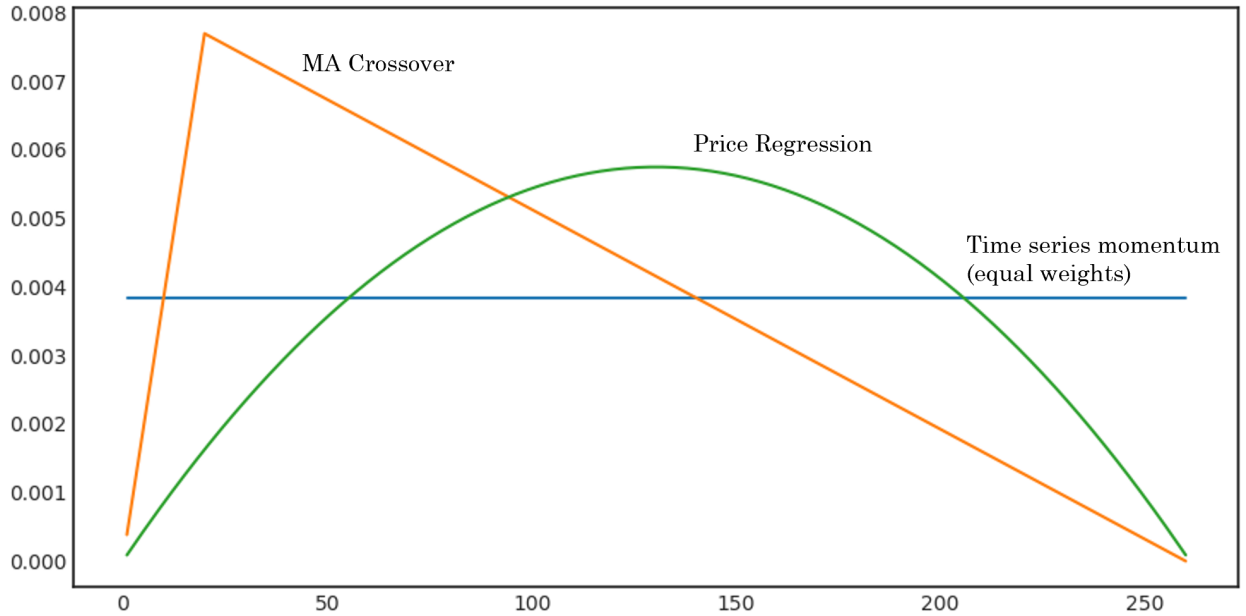


Figure 15: Trend Signature Plot 3

In the comparison in this and the previous sections, we deliberately set the parameters of different trend calculation methods to the same time scale (only such a comparison is meaningful). This is very important as it indicates that once the time scale is chosen, the effectiveness of different trend calculation methods is very close. Ultimately, these trend calculation methods can all be converted into their respective trend signature plots. Although their forms may differ, they are essentially similar weighted averages of returns.

4.3 An Example

This section uses the example from Levine and Pedersen (2016) to compare the TSMOM and MACROSS methods and to confirm that, for trend strategies, the time scale for calculating the trend is far more important than the specific method chosen at a given scale.

Specifically, for these two methods, consider three time scales: one month, three months, and one year. The investment targets are the 58 assets mentioned in Moskowitz et al. (2012) (including commodity futures, bonds, stocks, and forex). For these three time scales, the parameters for TSMOM and MACROSS are summarized in Table 1. The fast and slow moving averages in

Table 1: Trend Following Strategies at Different Time Scales

Time Scale	TSMOM	MACROSS
One Month	TSMOM(22)	MACROSS(3, 12)
Three Months	TSMOM(66)	MACROSS(8, 32)
One Year	TSMOM(260)	MACROSS(32, 128)

Table 2: Performance of Trend Following Strategies

Signal Name	Annual Return	Annualized Volatility	Sharpe Ratio
MACROSS(3,12)	10.3%	10.2%	1.01
MACROSS(8,32)	10.9%	10.3%	1.06
MACROSS(32,128)	12.8%	9.7%	1.33
TSMOM(22)	9.8%	10.1%	0.97
TSMOM(66)	12.1%	10.1%	1.20
TSMOM(260)	14.2%	9.8%	1.45

The data in this table are reproduced from Levine and Pedersen (2016).

MACROSS use exponential moving averages, with the parameters representing the centers of mass.

The returns, volatility, and Sharpe ratios of the trend following strategies constructed using these methods are shown in Table 2. They suggest that for a given time scale, the performances of TSMOM and MACROSS are very close; choosing the correct time scale for calculating the trend helps improve the strategy’s returns and Sharpe ratios. The key to the success of a trend following strategy lies in selecting the appropriate time scale, position control, portfolio construction, and risk management. Compared to these factors, the specific indicator or method used to calculate the trend is less important.

5 Position Sizing

Position sizing, also referred to as leverage management, is a critical component of quantitative strategies. One common leverage management method is the target volatility (or target risk) approach, which dynamically adjusts leverage to maintain portfolio risk at a predefined level. The leverage L is determined as:

$$L = \frac{\sigma_{\text{tgt}}}{\sigma_t^p}, \quad (11)$$

where σ_{tgt} is the target volatility level, and σ_t^p is the estimated volatility of portfolio p at time t . For example, if the target volatility is 10% and the current portfolio volatility is 5%, the leverage

is set to 2; if portfolio volatility rises to 20%, leverage is reduced to 0.5 (i.e., 50% position sizing).

Empirically, Perchet et al. (2016) demonstrate the efficacy of target volatility strategies across assets such as emerging market equities, the Russell 1000, S&P indices, and U.S. high-yield bonds, showing significant improvements in Sharpe ratios. Papageorgiou et al. (2017) find that volatility timing based on historical averages enhances annual returns by approximately 200 basis points while reducing maximum drawdowns in U.S. and Australian equity indices. The popularity of target volatility models is further evidenced by indices such as the S&P Dow Jones Target Risk series.

For trend following portfolios, Moskowitz et al. (2012) adopt a single-asset target volatility of 40%, but this approach neglects dynamic cross-asset correlations, leading to unstable realized portfolio volatility. Baltas (2015) emphasizes that target volatility should instead be applied at the portfolio level to ensure effective risk control. Let σ_t^{TF} denote the volatility of a trend following portfolio. The strategy can be formalized as:

$$r_{t,t+1}^{\text{TF}} = \frac{\sigma_{\text{tgt}}}{\sigma_t^{\text{TF}}} \sum_{s=1}^{S_t} \omega_t^s r_{t,t+1}^s, \quad (12)$$

where $r_{t,t+1}^s$ is the return of the trend following signal for asset s , and ω_t^s is its weight. Target volatility models provide a dynamic framework for adjusting leverage in trend following strategies, aligning portfolio risk-return profiles with investor preferences.

6 The Essence of A Trend Signal

Trend following is a right-side trading strategy that uses a trend measure to describe the strength of an upward or downward trend over a past period, assuming that the trend will continue. Regardless of the trend measure used, we cannot predict when a trend will end in advance, so trend strategies often incur losses when the trend reverses. This risk-return characteristic is inherent in trend strategies.

In this section, we attempt to answer why trend strategies are inherently right-side trading strategies and why trend signals have a (significant) lag. This explanation begins with the discussion about the essence of trend signals, using moving averages as an example. Mathematically, the essence of moving averages is a low-pass filter. Its purpose is to filter out high-frequency disturbances (noise) in a time series while retaining useful low-frequency trends. How can we extract the true low-frequency trend from a time series? The following mathematical explanation reveals the cause of the lag in trend signals.

Assume we have a time series $\{\dots, r_{t-2}, r_{t-1}, r_t, r_{t+1}, r_{t+2}, \dots\}$, as shown in Figure 16. Additionally, assume we have a filtering function $f()$ that operates on the time domain t (the specific form of f is determined by the chosen moving average algorithm).

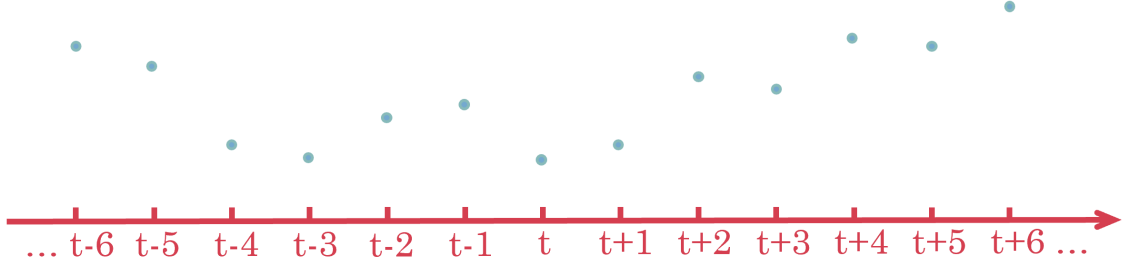
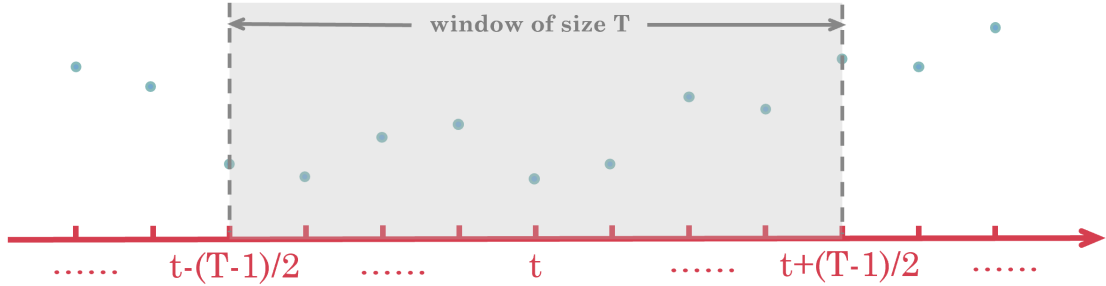


Figure 16: A Time Series Example

Theoretically, the low-frequency filtering at any time t (denoted as x_t) can be mathematically represented as the convolution of the time series $\{r_t\}$ with the filtering function f over the entire time domain:

$$x_t = \sum_{i=-\infty}^{\infty} f_i r_{t-i}. \quad (13)$$

where f_i represents the value of the filtering function at time $t - i$ (which can be understood as the weight of r_{t-i} ; for example, simple averaging uses equal weights). In practice, a window of length T is usually used to replace this infinite series, meaning that the filtering function f is only effective within the window and is zero outside the window, as shown in Figure 17.

Figure 17: Apply Window of Size T

With the window of length T added, the low-frequency filtering at time t becomes the convolution of the time series $\{r_t\}$ with the filtering function f within this window:

$$x_t = \sum_{i=t-(T-1)/2}^{t+(T-1)/2} f_i r_{t-i}. \quad (14)$$

The main problem with the above formula is that it uses future data to calculate the low-frequency component at time t . In other words, theoretical low-pass filtering (or moving smoothing) requires post-event data, assuming that all data points are known and the low-frequency components at all time points are calculated accordingly. However, this is impossible with real-time because, at any time t , we do not have future data. Since we cannot use data from $t - (T - 1)/2$ to $t + (T - 1)/2$ in real-time, we can only shift the entire window left by $(T - 1)/2$ units on the time axis and use

data from $t - (T - 1)$ to t to calculate x_t , as shown in Figure 18

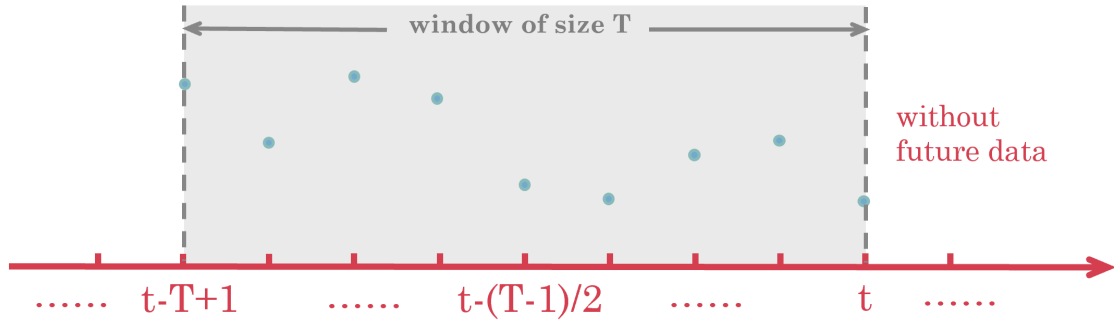


Figure 18: Apply Window of Size T without Future Data

After this adjustment, the low-frequency filtering of real-time data at time t becomes the convolution of the time series with the filtering function f from $t - T + 1$ to t :

$$x_t = \sum_{i=t-T+1}^t f_i r_{t-i}. \quad (15)$$

Taking the simple moving average as an example, within the window T , the value of the filtering function at each point is $1/T$ (equal weights). The result obtained using the above formula actually represents the low-frequency component at time $t - (T - 1)/2$, not at time t , but we have to use it as the low-frequency component at time t , thus introducing a lag of $(T - 1)/2$.

No access to future data in real-time (of course!) is the fundamental reason for the lag in trend signals.

From Eq. (15), we can see that the larger the time window T for calculating moving averages, the greater the lag $(T - 1)/2$ (for example, in stock data, daily moving averages such as MA20, MA30, MA50, etc., show more lag as the window T increases). Due to the relationship between lag and the time window for calculating moving averages, people instinctively resist using longer windows to calculate trends. Additionally, people prefer to use exponential weighting or other complex weighting methods to capture trend changes more quickly¹⁰, but the lag caused by the lack of future data makes trend following inherently a right-side trading strategy.

7 Properties of Trend-Following Strategies

To stick to trend following strategies, it is crucial for one to have a deep understanding of its risk and return properties and therefore to set correct expectations for its performance, which is the topic of this section.

¹⁰The financial industry has developed many different moving average algorithms for this purpose.

7.1 Low Sharpe Ratio

Moskowitz et al. (2012) calculate the Sharpe ratios of trend following strategies using a 12-month trend measure for different asset classes. Although trend following strategies achieve positive Sharpe ratios across all assets, most of the annualized Sharpe ratios are below 0.4. From the mainstream (yet irrational) market perspective, such performance is unattractive, but Sharpe ratios in this range are representative of the performance of trend following strategies on individual assets.

While there is limited room for improvement in trend following strategies on individual assets, the advantage lies in the low correlation among different asset classes. Thus, we can combine strategies on different assets to achieve diversification and improve risk-return characteristics¹¹.

7.2 Crisis Alpha

CTA funds employing trend following strategies gained prominence during the 2008 financial crisis for their exceptional performance, earning the reputation of a drawdown protector. Correlation analysis reveals that trend following strategies exhibit a low correlation (-0.09) with the Chinese stock market, providing effective diversification benefits. Beyond diversification, these strategies excel during market downturns, a feature termed “crisis alpha.” Kaminski (2011) defines crisis alpha as “Profits gained by exploiting persistent trends that occur across markets during times of crisis.”

During panic-driven sell-offs, most assets face liquidation, and market sentiment plummets. Trend-following strategies profit by shorting declining markets. Hamill et al. (2016) analyze trend following performance during equity and bond market crashes. By sorting quarterly equity and bond returns into quintiles, they demonstrate that trend following generates significantly higher returns in the worst-performing quintiles.

Hurst et al. (2017) evaluate trend following performance during the ten largest U.S. equity market crises, including the 2008 global financial crisis, the dot-com bubble, and the 1987 crash. Trend following delivered positive returns in 8 out of 10 crises, with minimal losses in the remaining two. Allocating 20% of a 60/40 portfolio to trend following enhances Sharpe ratios and reduces drawdowns.

In China, trend following also exhibits crisis alpha properties. Figure 19 plots quarterly returns of the Wind All-A Index, CSI Treasury Bond Index, and Nanhua Commodity Index against trend following returns. The “time-series momentum smile” (Moskowitz et al. 2012) is evident: trend

¹¹In a simulated study, we used Winton’s *The Future* tool (<https://tools.winton.com/thefuture/>), which takes parameters like the Sharpe ratio and correlation of individual strategies as input and conveniently visualizes the combined effect of these individual strategies. We considered 20 individual trend following strategies, each with a Sharpe ratio of 0.4 and a correlation of 0.1 between strategies. The simulation results showed that the portfolio’s Sharpe ratio increased to 1.16, significantly higher than the 0.4 of individual strategies. The distribution of the composite strategy’s Sharpe ratio can be obtained by running more simulations.

following thrives during extreme market movements, whether bullish or bearish. Notably, during panic-driven declines, trend following profits from short positions, serving as an effective hedge.

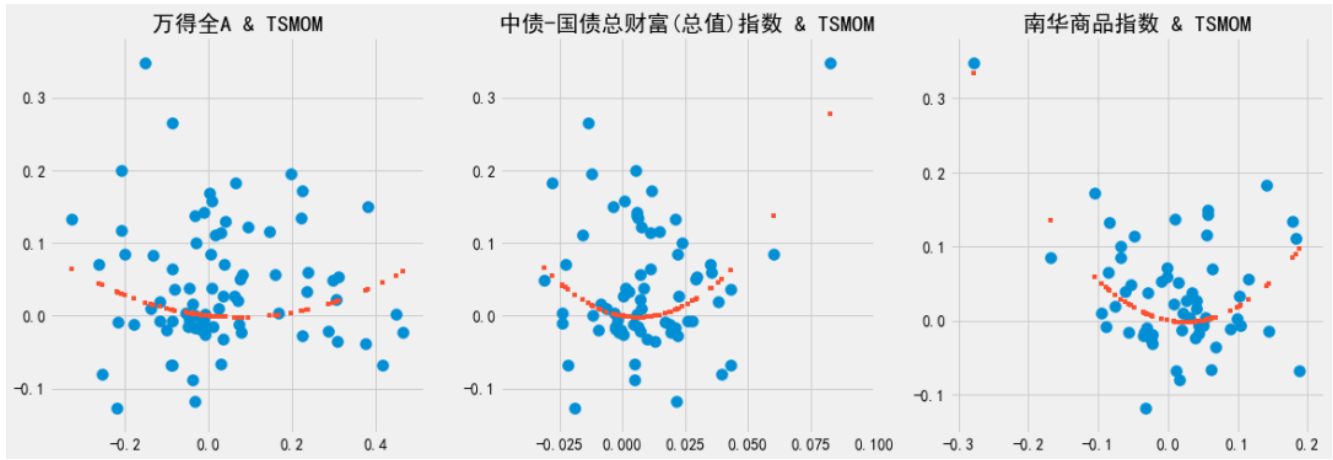


Figure 19: Time-Series Momentum Smile in China

To further assess drawdown protection, Figure 20 compares trend following performance during seven major drawdowns of the Chinese stock market, including the 2008 crisis, 2015 market crash, and 2018 prolonged decline. Trend following strategies generated positive returns in five episodes, with a remarkable 60% gain during the 2008 collapse. Even in negative episodes, losses were negligible relative to the broader market.

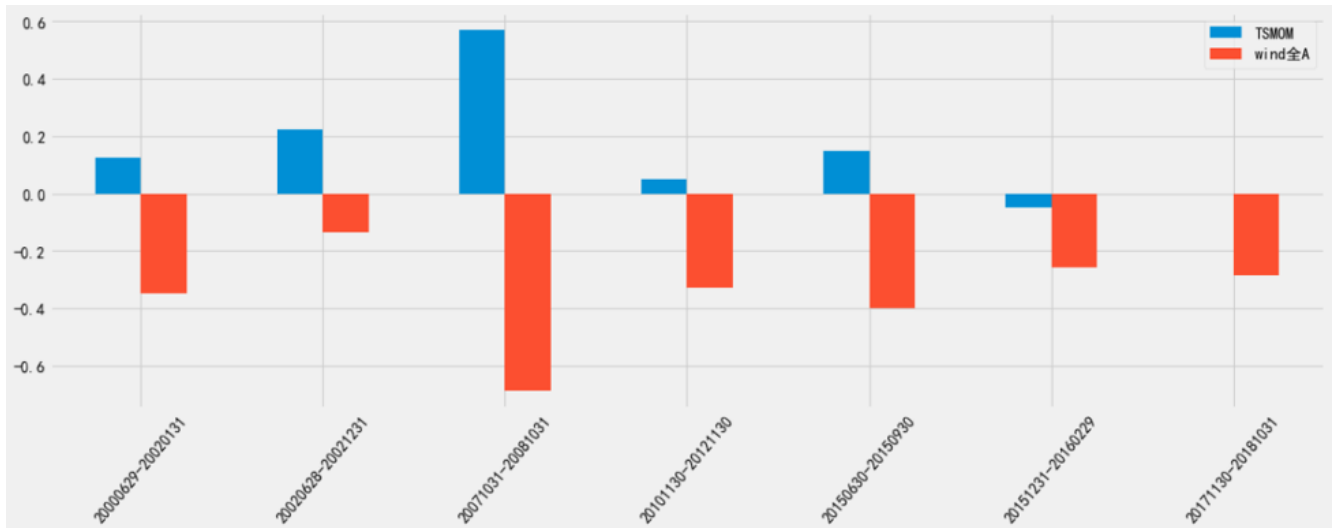


Figure 20: Trend-Following Performance During Drawdowns of the Chinese Stock Market

7.3 Long-Short Asymmetry

Similar to cross-sectional factor strategies, trend following exhibits long-short asymmetry in time-series signals. Zakamulin (2016) explores this asymmetry in S&P 500 timing strategies. While shorting during bearish signals (e.g., moving average crossovers) may enhance returns, holding cash

often yields higher Sharpe ratios due to lower volatility. Jordà et al. (2019) corroborate this by analyzing 150 years of global asset returns: equities' long-term upward drift (7.89% real returns globally) makes shorting suboptimal. Thus, “do not fight market beta” remains prudent advice.

7.4 Win Rate and Profit-Loss Ratio

Grinold (1989) posits that a strategy's information ratio (IR) depends on the information coefficient (IC) and breadth of bets (N):

$$\text{IR} = \text{IC} \times \sqrt{N}. \quad (16)$$

For trend following, analogous metrics include win rate and profit-loss ratio, which jointly determine strategy quality. To illustrate this, we consider 36 futures contracts in China. Figure 21 shows that trend following strategies for these assets achieve an average win rate slightly above 50%, with profit-loss ratios exceeding 1.0 for most assets, ensuring long-term profitability.

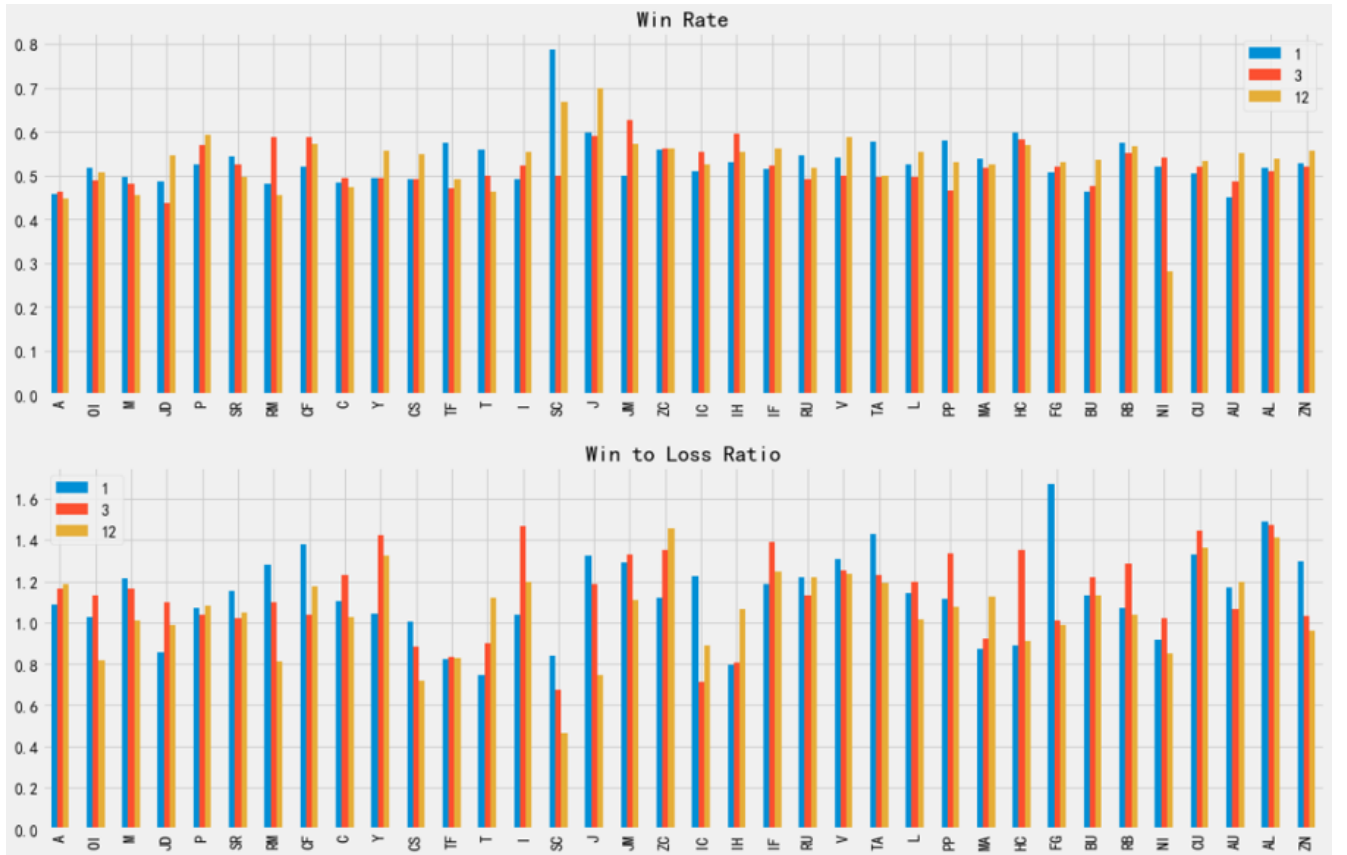


Figure 21: Win Rate and Profit-Loss Ratio

8 Empirical Evidence from China

Given the ubiquity and robust theoretical foundations of trend following strategies, this section further investigates whether such strategies exhibit comparable efficacy in China. Adopting the methodology of Moskowitz et al. (2012), we conduct an empirical study on Chinese futures markets.

8.1 Data

In selecting the investment universe, this study chooses 36 highly active assets, including three stock index futures, two bond futures, four metal futures, five energy and mineral futures, 10 industrial futures, and 12 agricultural futures. Given the limited number of futures contracts in the early years, the sample period begins on June 30, 1997. Since calculating trend following signals and variance-covariance matrices requires sufficient data, the actual empirical period starts on June 30, 1999, and ends on June 28, 2019, spanning exactly 20 years.

As China's futures market has developed and matured, the range of assets has expanded and diversified, ensuring the breadth of strategies is vast, offering ample opportunities for various approaches. Figure 22 illustrates the historical evolution of the number of futures assets. Initially, there were only three assets, which increased to four around 2000. From 2001 to 2004, the number of assets remained at five. Starting in 2007, the number began to grow rapidly, reaching 20 contracts by 2012. Between 2013 and 2016, the number further expanded, reaching 35. In recent years, new futures assets have been steadily introduced, and as of the sample end date, there were 36 assets.

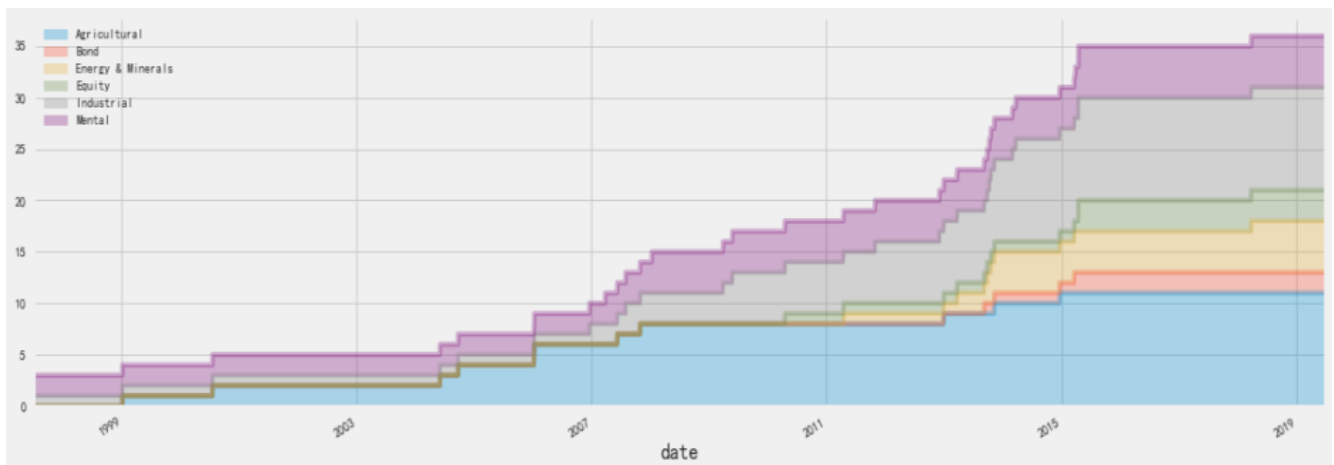


Figure 22: Historical Changes in the Number of Futures Assets in China

Table 3 presents the descriptive statistics of all these assets, including the start dates, the sectors they belong to, and monthly return distributions. From this table, it is evident that aluminum (AL), copper (CU), and rubber (RU) are the assets with the longest data histories, having been listed before the sample period began. Equity assets started having data after the

launch of the CSI 300 stock index futures (IF) in April 2010, while government bond futures data began after the launch of the 5-year government bond futures (TF) in September 2013. In terms of skewness, the left and right skewness are approximately equally distributed. Regarding kurtosis, almost all assets exhibit fat-tail characteristics.

Table 3: Descriptive Statistics of Futures Used in the Empirical Study

Asset	Count	Mean	Stdev	Min	25%	50%	75%	Max	Start Date	Skewness	Kurtosis	Sector
A	245	0.15%	4.87%	-15.80%	-2.57%	-0.36%	2.93%	16.09%	1999-02-26	0.2416	1.5082	Agricultural
C	177	0.15%	3.03%	-8.54%	-1.57%	-0.08%	2.10%	8.82%	2004-10-29	0.0033	0.4772	Agricultural
CF	180	-0.19%	5.33%	-13.49%	-2.72%	-0.74%	2.04%	25.90%	2004-07-30	1.6672	6.3892	Agricultural
CS	54	0.08%	4.74%	-10.67%	-3.13%	-0.32%	2.70%	12.02%	2015-01-30	0.3210	0.0569	Agricultural
JD	67	-0.25%	5.99%	-14.12%	-4.20%	-0.92%	3.47%	19.14%	2013-12-31	0.5016	0.8227	Agricultural
M	227	0.69%	6.26%	-18.41%	-3.52%	0.35%	4.11%	19.44%	2000-08-31	0.2591	0.4592	Agricultural
OI	144	-0.05%	5.94%	-22.47%	-2.61%	0.53%	2.97%	36.89%	2007-07-31	0.9004	11.147	Agricultural
P	140	-0.62%	6.62%	-30.49%	-4.53%	-0.70%	3.23%	24.53%	2007-11-30	-0.3576	3.6449	Agricultural
RM	78	0.84%	6.21%	-14.36%	-3.13%	0.78%	4.45%	19.08%	2013-01-31	0.3276	0.4958	Agricultural
SR	161	-0.36%	5.50%	-17.52%	-3.49%	-0.82%	2.40%	18.12%	2006-02-28	0.1843	1.5880	Agricultural
Y	161	-0.06%	5.83%	-21.62%	-3.55%	0.45%	2.49%	26.98%	2006-02-28	0.1417	3.7670	Agricultural
T	51	0.18%	1.07%	-3.10%	-0.34%	0.36%	0.93%	2.41%	2015-04-30	-0.9686	1.2605	Bond
TF	69	0.05%	0.86%	-2.97%	-0.50%	0.24%	0.54%	1.61%	2013-10-31	-1.0240	1.6666	Bond
I	68	1.70%	10.85%	-25.54%	-4.94%	1.95%	7.68%	24.42%	2013-11-29	0.0266	0.0161	Energy/Minerals
J	98	0.44%	10.48%	-23.08%	-6.29%	-0.86%	5.47%	39.77%	2011-05-31	1.0917	2.6716	Energy/Minerals
JM	75	0.68%	8.85%	-20.12%	-4.21%	-0.21%	3.20%	29.16%	2013-04-26	0.9491	1.8565	Energy/Minerals
SC	15	0.71%	9.10%	-23.45%	-3.76%	3.72%	6.29%	13.85%	2018-04-27	-1.3586	2.4938	Energy/Minerals
ZC	69	0.75%	5.79%	-10.08%	-3.45%	0.04%	5.07%	16.65%	2013-10-31	0.2710	-0.2434	Energy/Minerals
IC	50	0.84%	8.91%	-24.08%	-3.27%	1.52%	4.03%	23.13%	2015-05-29	0.1505	1.8193	Equity
IF	110	0.62%	7.28%	-18.69%	-4.58%	0.67%	4.43%	23.39%	2010-05-31	0.2310	0.8228	Equity
IH	50	0.41%	6.49%	-16.73%	-2.68%	0.98%	4.22%	15.19%	2015-05-29	-0.4643	0.3755	Equity
BU	68	-0.87%	8.01%	-21.72%	-6.56%	-0.18%	3.27%	22.89%	2013-11-29	0.3925	0.8998	Industrial
FG	78	0.42%	4.38%	-9.17%	-2.22%	0.42%	2.37%	12.07%	2013-01-31	0.2156	0.2276	Industrial
HC	63	1.28%	7.78%	-19.41%	-3.69%	-0.51%	6.55%	21.38%	2014-04-30	0.2260	0.1261	Industrial
L	142	0.28%	7.16%	-44.01%	-3.17%	0.51%	4.57%	23.63%	2007-09-28	-1.4807	10.035	Industrial
MA	92	-0.11%	6.63%	-22.56%	-3.69%	-0.48%	4.17%	18.46%	2011-11-30	-0.2415	1.5068	Industrial
PP	63	0.81%	6.31%	-11.17%	-3.53%	1.18%	3.22%	20.18%	2014-04-30	0.8000	1.4491	Industrial
RB	123	0.26%	6.93%	-23.33%	-3.86%	-0.33%	3.20%	20.79%	2009-04-30	0.3841	1.4957	Industrial
RU	263	-0.09%	8.54%	-29.92%	-5.36%	-0.88%	5.95%	31.38%	1997-08-29	0.1584	0.8124	Industrial
TA	150	-0.13%	7.29%	-34.47%	-4.38%	-0.49%	3.72%	28.99%	2007-01-31	-0.0440	3.9101	Industrial
V	121	-0.14%	5.12%	-12.78%	-3.12%	-0.69%	2.25%	22.59%	2009-06-30	0.7562	2.9211	Industrial
AL	263	-0.11%	4.03%	-12.55%	-2.32%	-0.30%	2.17%	15.28%	1997-08-29	0.2654	1.7514	Metal
AU	137	0.28%	4.98%	-17.93%	-1.93%	-0.11%	3.41%	13.66%	2008-02-29	-0.2256	1.3034	Metal
CU	263	0.61%	6.86%	-40.85%	-3.17%	0.40%	3.82%	35.26%	1997-08-29	-0.1992	7.1003	Metal
NI	51	0.14%	6.70%	-11.05%	-5.80%	-1.15%	5.50%	12.28%	2015-04-30	0.2396	-1.1925	Metal
ZN	147	-0.07%	6.92%	-37.22%	-3.18%	0.28%	4.08%	16.77%	2007-04-30	-1.2624	5.7459	Metal

8.2 Buy and Hold

The long-term performance of individual assets and sector portfolios (constructed using equal-weighted constituents) is detailed in Tables 4 and 5, respectively. First, annualized returns demonstrate substantial cross-sectional variation: 18 assets exhibit negative returns while 18 show positive performance. Bitumen (BU) records the lowest annualized return at -13.56% , whereas iron ore (I) achieves the highest return of 14.01% . Treasury bond and equity futures deliver positive returns.

Second, volatility characteristics reveal pronounced differences across asset classes. Equity, industrial commodities, and energy/chemicals sectors display the highest annualized volatility, fol-

lowed by metals and agricultural commodities, with treasury futures exhibiting minimal volatility. Regarding maximum drawdowns, all assets except bond futures experienced substantial historical declines, suggesting concentrated exposure to single assets could lead to significant losses.

Finally, focusing exclusively on commodity futures, individual assets show near-zero average and median annualized returns. However, a diversified and rebalanced long only portfolio generates statistically significant positive returns, consistent with Gorton and Rouwenhorst (2006).

Table 4: Individual Asset Performance

Asset	Return	Volatility	Cumulative Return	Maximum Drawdown
A	1.49%	16.80%	34.54%	-42.41%
AL	-2.25%	14.27%	-35.98%	-63.98%
AU	1.64%	17.25%	23.74%	-45.31%
BU	-13.56%	27.76%	-55.59%	-68.41%
C	1.11%	10.50%	19.29%	-29.53%
CF	-3.68%	18.46%	-44.04%	-59.88%
CS	-0.91%	16.43%	-1.71%	-29.37%
CU	7.39%	24.22%	326.0%	-61.11%
FG	2.76%	15.17%	29.07%	-37.36%
HC	12.91%	26.96%	85.51%	-48.69%
I	14.01%	37.60%	114.1%	-57.50%
IC	0.38%	30.85%	25.11%	-36.59%
IF	5.73%	25.22%	48.02%	-44.38%
IH	3.50%	22.48%	10.41%	-32.00%
J	-1.11%	36.31%	-7.32%	-81.65%
JD	-4.08%	20.77%	-24.96%	-51.05%
JM	5.21%	30.67%	26.08%	-55.23%
L	-0.05%	24.80%	-0.50%	-58.68%
M	6.34%	21.67%	211.23%	-39.68%
MA	-2.30%	22.97%	-26.40%	-41.85%
NI	-3.45%	23.22%	-3.94%	-38.59%
OI	-3.14%	20.58%	-26.99%	-64.27%
P	-9.67%	22.93%	-69.53%	-78.65%
PP	7.34%	21.86%	47.28%	-25.58%
RB	0.27%	24.01%	2.95%	-78.17%
RM	6.76%	21.53%	65.85%	-28.84%
RU	-5.04%	29.93%	-65.67%	-88.20%
SC	-0.90%	31.54%	-4.50%	-32.82%
SR	-4.67%	19.05%	-55.98%	-60.01%
T	1.55%	3.71%	9.34%	-6.24%
TA	-4.08%	25.24%	-44.75%	-58.25%
TF	0.65%	3.00%	3.02%	-4.61%
V	-3.71%	17.75%	-27.50%	-62.55%
Y	-2.87%	20.21%	-30.63%	-75.06%
ZC	7.32%	20.07%	50.17%	-51.73%
ZN	-4.73%	23.98%	-38.57%	-72.18%

Table 5: Sector Portfolio Performance

	Agricultural	Metal	Industrial	Energy/Minerals	Equity	Bond	All	All Commodity
Annual Return	1.65%	3.24%	0.18%	-2.39%	5.71%	1.08%	2.53%	2.45%
Annual Volatility	15.58%	16.58%	24.15%	27.75%	25.37%	3.33%	14.12%	14.36%
Cumulative Return	38.61%	99.35%	0.06%	-16.69%	47.75%	5.57%	64.78%	62.33%
Maximum Drawdown	-42.32%	-52.52%	-73.98%	-78.72%	-44.38%	-5.42%	-45.24%	-47.95%

8.3 Return Persistence

One of the key points to profitability in trend following is the persistence of returns. Generally, the better the return persistence, the smoother the trend, and the greater the potential profits for the strategy. Before constructing the trend following portfolio, we first explore the persistence of returns across different futures assets.

The method used is to conduct a panel regression, where the risk-adjusted monthly returns of all assets are the dependent variable, and the lagged risk-adjusted monthly returns (lagged h months) are independent variables. The significance of the regression coefficients can be used to measure return persistence. The panel regression model is:

$$\left(\frac{r_{t,s}}{\sigma_{t,s}}\right) = \alpha + \beta_h \left(\frac{r_{t-h,s}}{\sigma_{t-h,s}}\right) + \epsilon, \quad (17)$$

where $r_{t,s}$ is the return of asset s in month t , $\sigma_{t,s}$ is the realized volatility of asset s at time t , and h is the number of months lagged.

The lag period h ranges from 1 to 60 months. For each lag h , a panel regression model is estimated, and the t -statistic for the regression coefficient is obtained. Intuitively, if the regression coefficient β_h is significantly greater than 0, it suggests good return persistence; if β_h is significantly less than 0, it suggests clear reversal characteristics. The reason for using risk-adjusted returns is that volatility varies greatly across assets, and using returns without adjustment would lead to incomparable results. The realized volatility for each asset is calculated using an exponentially weighted method based on daily returns.

Figure 23 plots the t -statistics for all lags under consideration. It can be observed that for lags between 1 and 12 months, the regression coefficients are significantly greater than 0 most of the time, indicating high return persistence within one year. On the other hand, beyond 12 months, a reversal pattern becomes more evident, which aligns with the observations of Moskowitz et al. (2012).

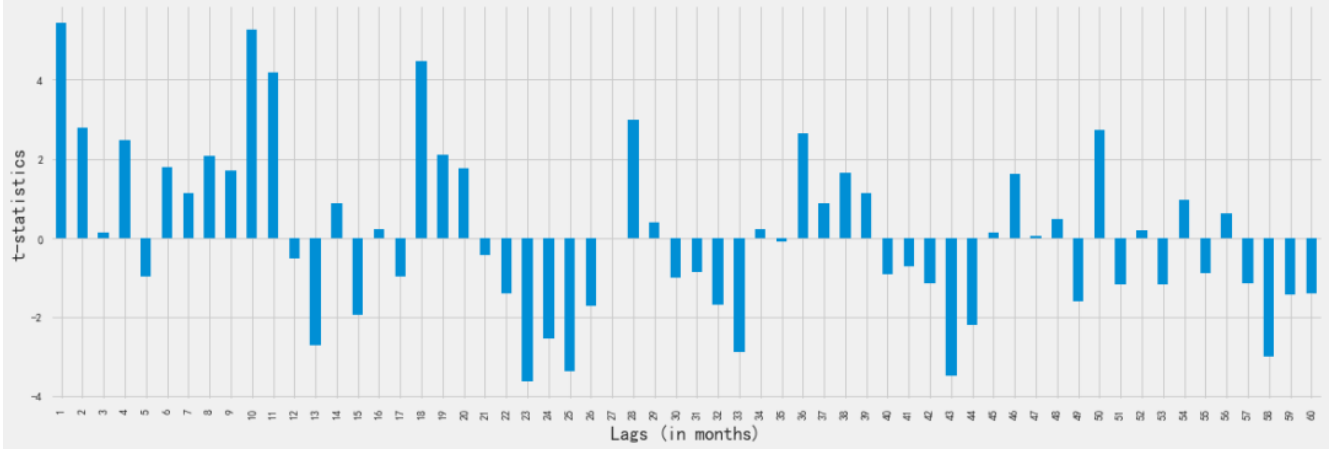


Figure 23: Return Persistence Analysis

8.4 Portfolio Construction

Next, following the methods of Moskowitz et al. (2012) and Hurst et al. (2013), we construct trend following portfolios. At the end of each month t , for each asset s , we calculate its return over the past h months. The direction of the trade is determined based on the sign of this return: if positive, the position is long; if negative, the position is short. Specifically, the trend following return for asset s in month $t + 1$ is:

$$r_{t,t+1}^{TF,s} = \text{sign}(r_{t-h,t}^s) r_{t,t+1}^s, \quad (18)$$

where $r_{t,t+1}^s$ is the return of asset s in month $t + 1$, $r_{t-h,t}^s$ is the cumulative return of asset s over the previous h months, and h is the lookback window, with values of 1, 3, and 12 corresponding to 1 month, 3 months, and 1 year.

The weight of each asset is determined by the inverse of its volatility. This volatility-inverse weighting, also known as naive Risk Parity, is a simplified version of risk parity that considers only the volatility of the individual components, not their correlations. Assets with higher volatility (such as stock index futures) will have smaller weights, while assets with lower volatility (such as bond futures) will have larger weights. Inverse volatility weighting helps ensure that the risk contribution of each asset is balanced, thus providing diversification in the portfolio's risk sources.

The leverage for each asset is determined by a fixed target risk, set at an annualized volatility of 40%. This means the leverage for each asset is adjusted based on its volatility level to achieve the target risk. The reason for setting the target at 40% is not arbitrary, but to ensure that the overall trend following portfolio has an annualized volatility of approximately 12%, which aligns with classical factor models, such as Fama and French (1993)'s three-factor model. After considering the weights and leverage, the trend following return for asset s in month $t + 1$ is:

$$r_{t,t+1}^{TF,s} = \text{sign}(r_{t-h,t}^s) \frac{40\%}{\sigma_t^s} r_{t,t+1}^s. \quad (19)$$

For each asset, we compute its performance over three different periods and combine the results using equal weights. The results are shown in Figure 24. It can be seen that 12 out of the 36 assets have a negative annualized return, accounting for one-third of all assets, with six of them being agricultural products. The best-performing sectors are energy and minerals and industrial products, with metals such as copper, aluminum, and zinc performing well. Agricultural products show greater variation, with some performing better than others. Among stock index futures, the CSI 300 has an annualized return exceeding 10%, while the other two have more average performances due to insufficient time length. Both government bond futures have negative returns.

From a risk perspective, all assets have realized volatilities above 20%, with three-quarters of the assets having volatilities above 30%, indicating that trend following strategies for individual contracts carry significant risk. However, the majority of assets have volatilities below 40%, which is below the target risk level, indicating that multi-period allocation provides a diversification effect.

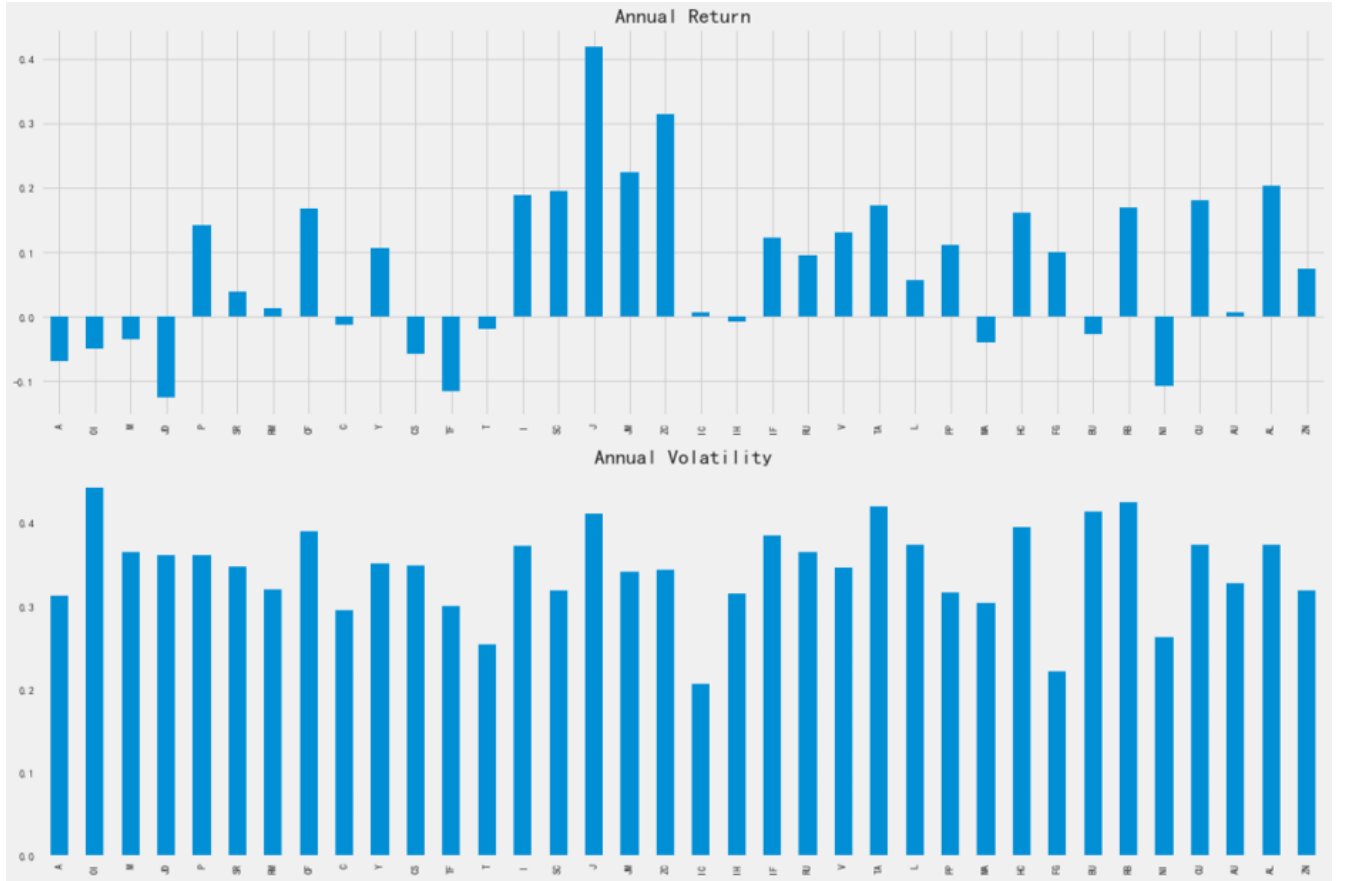


Figure 24: Trend-Following Performance of Individual Assets

Summing all the assets using the following formula results in the trend following portfolio, where S_t represents the number of assets at time t :

$$r_{t,t+1}^{TF} = \frac{1}{S_t} \sum_{s=1}^{S_t} \text{sign}(r_{t-h,t}^s) \frac{40\%}{\sigma_t^s} r_{t,t+1}^s. \quad (20)$$

Figure 25 and Table 6 show the performance of trend following strategies across three different trend signal calculation windows (i.e., 1, 3 and 12 months) and their equal-weight combination (labeled as TSMOM). The Buy-and-Hold strategy is included for comparison, with all assets weighted equally and rebalanced monthly. It can be seen that the overall trends of the three different periods are similar, although there are slight variations in the results. All strategies achieve high annualized returns, but also experience significant drawdowns. The best-performing window is the 3-month window. The combined three-period equal-weight portfolio has an annualized return of 16.24%, an annualized volatility of 18.51%, and a maximum drawdown of -22.63% , all of which outperform individual periods. When compared to the equal-weight Buy-and-Hold strategy, the trend following strategy shows a clear advantage, with a Sharpe ratio of 0.88, compared to the Buy-and-Hold Sharpe ratio of 0.18.

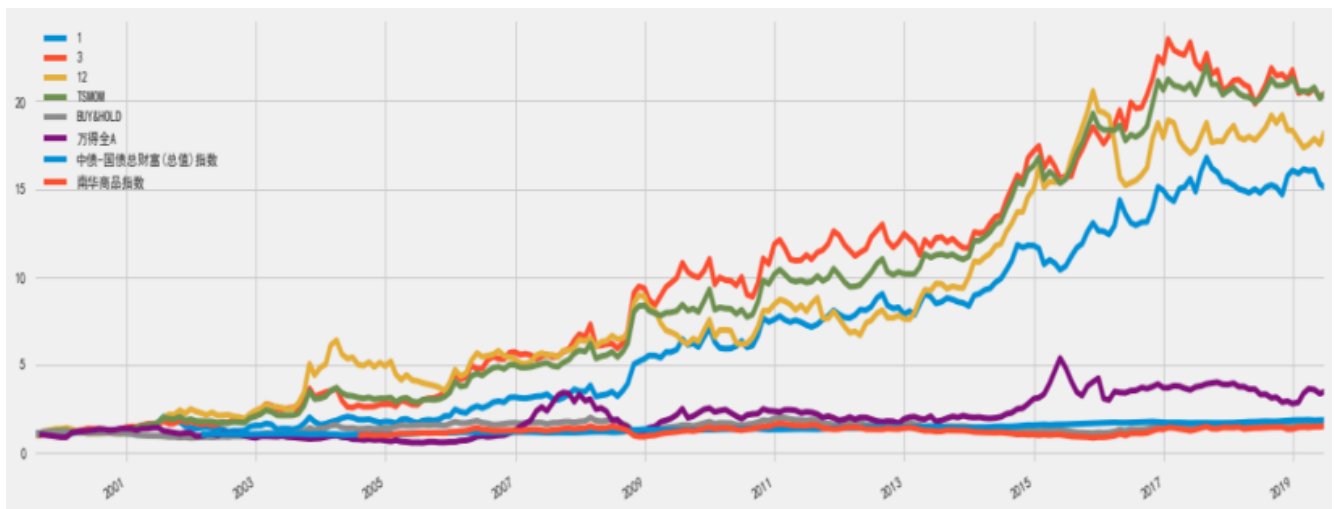


Figure 25: Performance of Trend Following Strategies in China

Table 6: Risk-Return Analysis of Trend Following Strategies

	1-month	3-month	12-month	TSMOM	Buy-and-Hold
Annual Return	14.32%	16.23%	15.80%	16.24%	2.53%
Annual Volatility	20.90%	20.95%	24.15%	18.51%	14.12%
Maximum Drawdown	-42.42%	-30.29%	-45.26%	-22.63%	-45.24%
Sharpe Ratio	0.69	0.77	0.65	0.88	0.18

Finally, when compared to common asset classes, trend following demonstrates its distinct appeal. Correlation analysis shows that, compared to other assets, trend following has a negative correlation with stock market and a near-zero correlation with bond market, allowing for an ideal diversification configuration.

9 Has It Failed?

In general, combining trend following strategies across multiple assets can (largely) improve risk-adjusted return, i.e., the Sharpe ratio. However, we must also recognize that multi-asset trend following strategies can still incur sustained losses in the short to medium term (i.e., 1 to 3 years). If one looks at the performance of the famous Barclay CTA Index¹², it is clear that its performance since July 2011 has been rather dismal, with almost continuous drawdowns since 2015.

Depending on the market conditions, trend following strategies can have periods of good performance and prolonged periods of losses. In the long run, trend following strategies can yield excellent returns and are worth allocating to. However, those who can truly allocate and manage it must have a very clear understanding of how poor its performance can be in the short to medium term. We next discuss this issue from three aspects, i.e., strategy capacity, asset correlation, and lack of trends.

9.1 Capacity

The success of trend following in 2008 brought it into the spotlight, resulting in a surge of attention and influx of capital. Baltas and Kosowski (2013) provide an overview of this phenomenon. By the end of 2011, CTA funds had surpassed 300 billion within the 2 trillion hedge fund industry. However, CTA performance was underwhelming, and one possible reason for this is that the capacity had reached a ceiling.

Baltas and Kosowski (2013) use two methods to explore whether market capacity affects trend following strategies. The first method was a regression analysis with fund inflows as the independent variable and trend following performance as the dependent variable. The results showed that the regression coefficient was negative but not statistically significant, indicating that fund inflows did not affect the performance of trend following models. The second method involved capacity estimation, assuming the entire AUM of systematic CTAs were invested in trend following strategies. The results showed that the positions required for most assets did not exceed the total position size of each asset; in fact, the positions of trend following models represented only a small fraction of the overall derivatives market. Therefore, the poor performance of trend following was not significantly related to capacity.

9.2 Increased Asset Correlation

Georgopoulou and Wang (2016) also investigate the poor performance of trend following in recent years, attributing the primary reason to a shift in asset correlations. Specifically, after the financial crisis, central banks worldwide adopted quantitative easing (QE) to stimulate economic

¹²Considering the high proportion of trend following in CTAs, we use the Index's performance to represent the historical performance of trend following strategies.

development, which disrupted and distorted the correlation among various asset classes. Unlike cross-sectional momentum, time-series momentum strategies prefer assets with low correlations. The increasing correlations led to fewer independent trends, weakening the effectiveness of trend following strategies.

To confirm this hypothesis, Georgopoulou and Wang (2016) perform a regression with trend following returns as the dependent variable and the QE dummy variable as the independent variable. The results showed that the QE coefficient was significantly negative, indicating that quantitative easing indeed affects trend following performance.

Additionally, Baltas (2015) and Baltas and Kosowski (2017) point out that the increase in asset correlations might be one of the reasons for the poor performance of trend following strategies. Baltas (2015) calculates the average correlation coefficient using intra-month data, then divided the assets into four groups based on the strength of their correlations, and evaluated the trend following performance for each group. The results showed that the higher the correlation, the lower the returns. Baltas and Kosowski (2017) also find that considering the correlation coefficient helps improve the performance of trend following strategies.

9.3 Lack of Trends

AQR's 2019 report (Babu et al. 2019) provides a detailed analysis of the reasons behind the failure of trend following strategies. First, they suggested that the lower risk-free rate after the financial crisis might be one of the factors contributing to the poor performance of trend following strategies (though not the main cause). This is easy to understand because futures are margin-based and do not require full capital allocation. Cash management can boost trend following returns, and a lower risk-free rate naturally reduces this boost.

Secondly, AQR argued that lack of diversification was not the reason behind the poor performance of trend following. To verify this, AQR defined a Diversification Multiplier, which is calculated by dividing the Sharpe ratio of the trend following portfolio by the average Sharpe ratio of individual trend following strategies. The larger this value, the better the diversification effect. Comparing the diversification multipliers for the full sample (1880–2018) and for 2010 to 2018, they found minimal difference, indicating that insufficient diversification should not be blamed.

Finally, AQR pointed the finger at the lack of trends. Their analysis revealed that between 2010 and 2018 the general trend of assets was weak, which was the main reason for the poor performance of trend following strategies. That said, AQR is not pessimistic about the future of trend following. If asset trends strengthen in the future, trend following strategies could perform significantly better.

10 Skepticism

Although trend following has proven itself over a long period, not everyone is an advocate of this strategy.

Kim et al. (2016) point out that time-series momentum only outperforms buy-and-hold and cross-sectional momentum strategies when volatility adjustments are applied. Without volatility adjustments, there is no significant difference between time-series momentum and the buy-and-hold strategy. Furthermore, even with volatility adjustments, the difference between time-series momentum and buy-and-hold strategies is not significant. They conclude that the reason time-series momentum is effective is because it involves the use of leverage.

In addition, Huang et al. (2020) publish a paper titled “Time-series momentum: Is it there?”, which critically examine Moskowitz et al. (2012). In particular, Huang et al. (2020) study the same assets as Moskowitz et al. (2012). The authors first tested the predictive power of raw returns of different assets for future returns and only eight (three) assets had significant predictive ability at the 10% (5%) significance level. Out-of-sample performance was even worse, with 45 assets having negative out-of-sample R-squared values. They then replicated the panel regression of Moskowitz et al. (2012), obtaining similar results, but pointed out that Moskowitz et al. (2012) did not include an intercept term in their regression, implying an important assumption that assets have the same Sharpe ratio. However, it is clear that stocks, commodities, foreign exchange, and bonds do not have the same Sharpe ratio.

Based on this, Huang et al. (2020) argue that it is necessary to include the average (volatility-adjusted) returns of different assets in the model to control for the fixed effects; otherwise, the estimation results would be upwardly biased. The authors applied two bootstrap methods for adjustment and found that after controlling for fixed effects, the lagged one-month returns generally had no significant predictive ability. Although cumulative returns over a period still exhibited significant predictive power, their significance declined sharply. Huang et al. (2020) further test out-of-sample performance and find that past returns have significant out-of-sample predictive power in the stock market, but not in other markets, especially in the bond and foreign exchange markets, where out-of-sample R-squared values are generally negative.

Overall, Huang et al. (2020) conclude that past returns do not have significant predictive power, suggesting that the foundation of time-series momentum, as proposed by Moskowitz et al. (2012), is not robust. They then proposed the TSH strategy, which does not require return predictions and determines trade direction based on the asset’s historical average return sign. The test results reveal that the equal-weighted time-series momentum and TSH strategies do not have significantly different performances, leading to the conclusion that time-series momentum might be driven by differences in historical returns between assets. In other words, the excess returns of time-series momentum are more likely due to the risk premiums earned from holding positions.

11 Conclusion

Trend following is widely present across various asset classes, and its underlying logic is well-founded. From a technical standpoint, the most important aspect of a successful trend strategy is determining the appropriate time scale for the trend. The parameters of the strategy should be adapted to the return characteristics of the asset. Once the time scale is established, the specific choice of trend measure does not make a fundamental difference. It might be more effective to focus efforts on position management to control risk.

From a risk perspective, trend following is considered a divergent risk-taking strategy. While it may not be a “holy grail,” its return distribution aligns with the “holy grail distribution,” which gives it advantages such as crisis alpha, making it highly valuable for portfolio allocation. However, it is worth noting that studies have shown that excessive use of leverage amplifies both the returns and risks of such strategies in a nonlinear way. Therefore, sound and effective risk management is a crucial component of utilizing trend following strategies.

Although the return distribution of trend following strategies shows positive skewness, from a psychological perspective, successful implementation of trend following strategies requires a clear understanding of their risk-return characteristics. The Sharpe ratio of trend strategies on individual assets is usually not high, but combining trend strategies of different low-correlation assets can significantly enhance the portfolio’s Sharpe ratio. Even so, trend following strategies are very likely to experience sustained drawdowns over shorter to medium-term periods (e.g., one year). Mastering trend following strategies is not about understanding their positive long-term expectation but rather about comprehending their risk-return characteristics and accepting their potential underperformance in the short term.

As an ancient yet highly resilient strategy, trend following strategies warrant our serious study and research.

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